

DOGS

GENERAL CARE AND MANAGEMENT

Although sometimes disregarded as too basic, good knowledgeable care and management will greatly lengthen your dog's life and make it healthy and full of enjoyment. A dog's basic needs are few, but if ignored, health problems and disease will follow. Likewise, a few dollars spent on vaccinations and routine veterinary care can pay big dividends throughout your dog's life.

Housing

A dog's housing requirements are few. While many adapt well to life indoors, they still appreciate attention to these minimal needs. And dogs required to live outdoors, such as guard and stock dogs, need additional housing attention as they are more exposed to the elements.

Bedding

Even if your house dog sleeps on the couch or on your bed, it should have a bed or a spot of its own. Here it can escape children, catch quick naps, and keep its bones and toys. The bed should be slightly bigger than the dog is when the animal is curled up. A larger bed is not as comfortable to the dog, because a dog likes to feel something around it while napping.

Doghouses

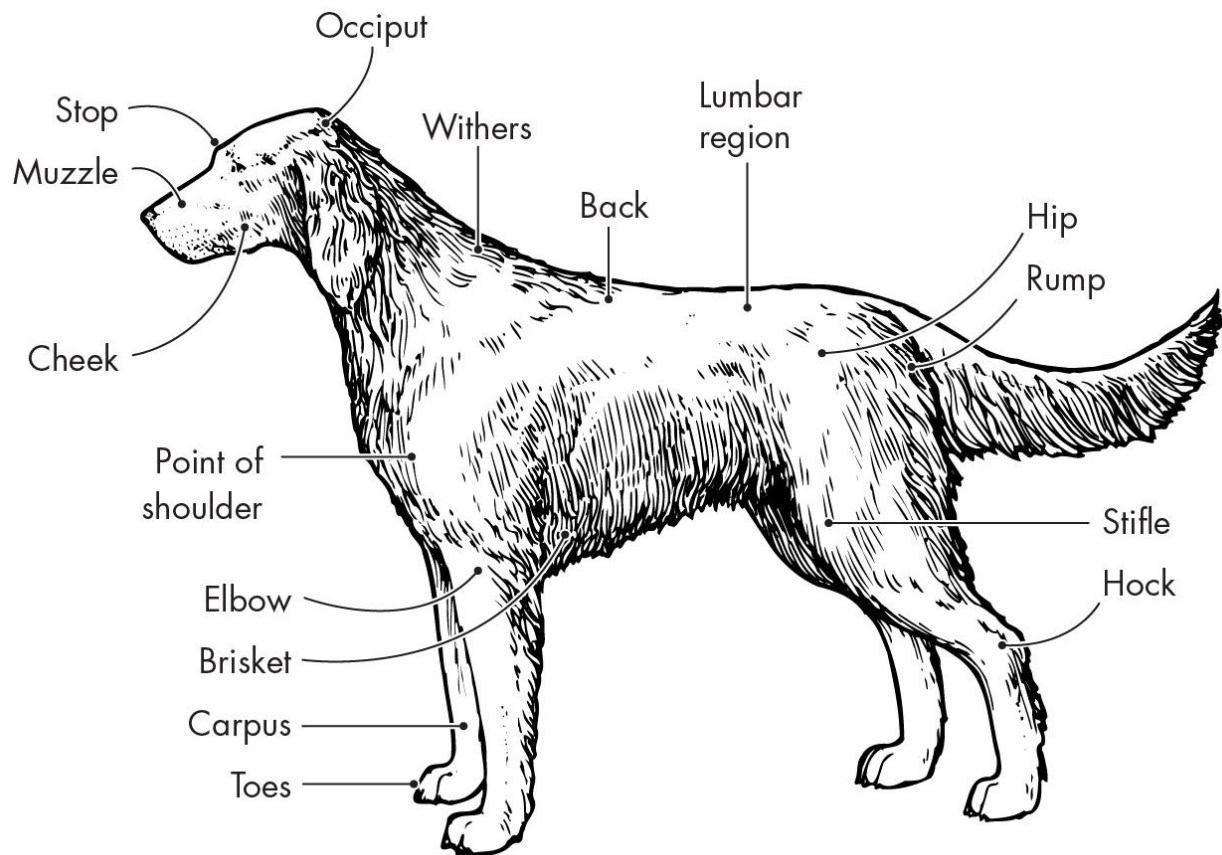
The outside dog needs a snug doghouse as protection from the elements. It should be solidly constructed, with a windbreak at the door. Except in very cold climates, it should not be insulated. Insulation sometimes allows the humidity to build up, causing condensation to form. This thoroughly chills the dog and makes it very susceptible to illness.

The house should face the south and be just a bit larger than the dog is when

the animal is curled up. This will let the dog's body heat warm the house in cold weather. The roof should be only an inch or two higher than the standing dog's back, for the same reason. Make the roof flat with a mild slope to it. Many dogs like to lie on top of their houses to sun themselves and get a better view.

There should be plenty of fresh bedding in the house in the winter as a dog likes to curl up in a "den." Even during the summer, some bedding is necessary to provide comfort during cool or rainy spells. Keep the bedding fresh and dry. After a rainy spell, change or dry the bedding, as necessary. Old blankets, deep cedar shavings, or a nice dog bed are all good, provided you keep them clean and dry.

PARTS OF A DOG



Move the doghouse from place to place periodically. This not only keeps the area cleaner and allows the grass to regrow but also aids in parasite control.

Dog Runs

A dog kept outside, such as a farm dog or hunting dog, also may need restraint, be it an enclosed dog run, a chain, or a fenced yard or pen. Unless you live in an isolated area, any dog must be kept restrained when unsupervised. Even farm dogs need restraint unless they are thoroughly trained to stay at home. Thousands of deer and other animals, including sheep, are killed by someone's pet out "having fun." Your dog doesn't have to be kept shut up all the time, just when there is no one around to keep an eye on it.

If you're building a closed-in run with a doghouse, use chain link or welded wire fence at least 6 feet high, and make the enclosure 4 × 12 feet. If your dog is a jumper, you may have to add a top. A cement floor is easiest to clean and keep parasite-free; it will also prevent the dog from digging. Pea gravel and crushed rock are other popular flooring materials. They are better on the dog's feet than cement but harder to keep clean. And a dog will often dig holes in the run, which makes it appear dirty. Place the doghouse at the end of the run, outside the enclosed area. Otherwise, the dog can climb up on the house and use it to jump over the fence.

First Aid Kit for Dogs

Here's a list of first aid items and supplies you'll want to keep around—just in case! Every dog owner should have at least the items marked with an asterisk. The other items can be useful if you own several dogs or if you are a breeder. Your veterinarian can make more suggestions based on your location and situation.

- Betadine
- Collar*
- Cotton*
- Ear mite remed*
- French
- French feeding tube
- Leash*

Pet nail clippers*
Puppy nursingbottle
Rectal veterinary thermometer*
Styptic pencil or blood
Styptic pencil or blood stopper*
Tweezers/forceps*
2-inch gauze*
Veterinarian's phone number*
Whelping nest heating pad

If you must chain your dog, use a chain heavy enough so that it will not break when the dog gets excited. Many dogs have been hung while dragging a length of chain after breaking loose. The chain should be 10 to 15 feet long, with a strong swivel at each end. Very strong dogs, such as Huskies, should have shorter chains. Otherwise, the dog may get a foot or its head in a loop of chain and then get excited and run off, breaking a leg or even choking itself. A dog that hits the end of its chain may have trouble with hematomas (large bruises) or throat injuries. Fastening the end of the chain to an old inner tube can provide enough "give" to prevent many of these injuries.

Feeding

Mature dogs in good health should be fed one meal a day or have free access to dry food at all times. The average dog, given food free-choice, will not overeat and will stay in just the right shape, provided it gets some regular exercise.

Whether fed free-choice or once daily, dry dog food of good quality is generally the best choice. It is better for the teeth than moist food since its abrasive quality keeps them clean and free of tartar buildup. Most well-known brands of dry food are a completely balanced diet, whereas many canned foods are not. An all-meat diet is not a balanced diet for a dog. In the wild, canines eat not only meat but also stomach contents, bones, and wild fruits and grasses. Whatever food you choose, read the list of ingredients. A lot of cheap dog foods have a very high cereal content, with very little meat or meat by-products. If

cereal grains are listed first, one after another, chances are very high that the food is composed largely of cereal. Look elsewhere for a good food.

To increase consumption of dry food, you can add chopped or ground raw meat to the dry food and mix it in well. (Some dogs need a bit more weight than they normally carry, and the added consumption of dry food will aid in this needed weight gain.)

In addition to dry food, your dog can also have meat scraps, cooked eggs, stale bread, cottage cheese, milk (in limited quantities), and table scraps as side dishes. Dogs should never receive more “goodies” than they do dry food, however, as this is what makes a picky eater. It is like feeding children chocolate cake and then asking them to eat their sandwich and vegetables.

Do not feed your dog highly spiced, fried, or greasy foods or foods with sharp bones, such as fish, poultry, or pork chops. Cooked bones are brittle, sharp, and dangerous, able to penetrate the bowels and stomach. Feeding potato chips to a dog can also be dangerous. Some dogs do not chew their food; and when unchewed potato chips reach the stomach, the sharp edges and points can cause so much pain that the dog can actually be thrown into convulsions. In fact, any indigestible food can produce gas or stomach pain, which in turn can cause convulsions.

Unless you are feeding plain dry food free-choice, throw away any food that's left by the next feeding, and give a little less at that feeding. You don't want the dog to always be hungry, but you do want it to finish its food between feedings. Never leave moist food, such as dry food mixed with liquid or canned food, in the dish all day. It can become sour or have high bacterial growth, and your dog may get sick if it gets hungry enough to eat the food later.

Like people, some dogs have certain foods that disagree with them and cause digestive upsets, from flatulence to diarrhea. Foods that occasionally cause problems include eggs, milk, soy products, and corn products. If your dog experiences digestive upsets, try changing its food to see if that helps.

Exercise

A house dog should receive exercise of some type every day. A tiny dog can get exercise by chasing a ball, playing tug-of-war, or just being naturally active. A larger dog should either be allowed to run in a fenced yard or be taken for a walk every day. A dog that receives insufficient exercise is apt to become fat and develop more physical defects, and it often becomes neurotic. Letting a dog

roam freely for its exercise is not a good idea. Not only is it dangerous for the dog, but it can cause trouble for you, as well. An unsupervised dog is like an unsupervised child: It can get into too much trouble without half trying! Tearing open garbage bags, digging in neighbors' flower beds, chasing bicycles, and bothering neighbors' pets are all common problems with unsupervised dogs.

A dog that is tied or kept in a run should have regular periods of free time or exercise out of confinement. Whether it is work, such as hunting or herding stock, or just a good run in the woods with its owner, the dog will appreciate it and will not become bored. A bored dog is often a barking dog.

Restraint

Most dogs are easy to restrain for examination and treatment as they are trusting of people and seem to know that they are trying to help. But sometimes a dog becomes frightened or is in severe pain, which transcends reason. In such cases safe and effective restraint is necessary.

First of all, a strong and safe collar is a must, if only to lead the dog into a veterinarian's office—where even a brave, calm dog fears to go. The collar should fit snugly enough that the dog cannot slip its head free by twisting and pulling back, and it should be loose enough for comfort. A flat nylon or leather collar is best. A choke collar can be used for everyday, but it is best to use it only as a training collar because a frightened dog can choke itself trying to escape in an emergency.

A strong, dependable leash is also necessary, if only from time to time, as in going to the veterinarian's clinic. The leash should be nylon or leather, and the snap should be heavy enough to hold the dog, whatever it decides to do.

No matter how well trained your dog is, use the leash and collar any time you have it in strange surroundings or if it has been injured. A frightened or injured dog may panic and run off, totally unaware of its owner's repeated calls.

To effectively restrain a small dog, place one hand on the back of its neck, over the collar (which can be grabbed, if needed). Place the other under its chin, and tip the head up slightly. Massaging the neck a bit and speaking reassuringly to it will also help.

To restrain a large dog, straddle its neck, and tip its head up with one hand while you hold its collar with the other. In this way, you can hold it while a helper takes its temperature, removes a burr from its tail, or performs whatever

minor treatment is necessary.

If additional restraint is necessary, such as when a smaller dog tries to nip the person holding or treating it, simply tie a length of 1-inch gauze snugly around the muzzle and then behind its ears in a bow. This works great for treating a small dog with a broken toenail, a minor cut that needs examination, or other minor problem.

Breeding

A female dog will come into heat roughly every 6 months. Smaller breeds usually begin cycling at 6 months of age, with larger breeds coming into their first heat between 8 and 12 months. Any female that you do not specifically plan to breed should be spayed before her first heat period. There are too many unwanted puppies out there to allow a chance breeding and therefore contribute to the immense problem already present. Spaying will not alter a female's personality or make her fat and inactive, and it will help prevent such things as uterine infections and breast tumors. And never breed a bitch unless you have the time, space, and money necessary to raise the puppies until you can find homes for them. It is no solution to say, "I'll just sell the extra ones to a pet shop." A high percentage of pet shop puppies never reach adulthood.

If you have a purebred female with a good disposition, conformation, and pedigree, wait until after her second heat period, at least, to breed her. It is not advisable to breed the bitch on her first heat as she is still a puppy herself then; she will not be able to handle the nutritional requirements of her unborn puppies and the nutrients necessary for her own growth at the same time. Thus, she may never reach the size and conformation she would have attained if you had waited until at least her second, or preferably her third, heat to breed her.

Choose the male, and make arrangements with the owner well in advance of the date your bitch is due to come into heat. Don't choose the male on bloodlines or looks alone; also consider his temperament. The most beautiful dog in the world is of no use if it is neurotic or mean. Also, take special care to avoid genetic or hereditary problems. Most breeds of dogs have an undesirable characteristic that dedicated breeders are trying to weed out, such as deafness in Dalmatians and hip dysplasia in German Shepherds. Never breed a dog with any hereditary defect or that has parents with a defect, as there is too much of a chance that the puppies will either pick it up, too, or pass it on to *their* puppies. Study your breed well, and make a good choice based on soundness, good disposition, and good conformation—not the popularity, length of hair, color, or

some such trait.

Before breeding, the female should be in good physical condition and free of worms, skin parasites, and vaginal or uterine infections. She should also be up-to-date on all her vaccinations. She will be in heat for approximately three weeks but will generally accept the male only from the 10th to the 14th day. Take her to the male early so that she can become accustomed to him and his surroundings. A frightened bitch may not accept the male and may even injure him.

After a successful breeding, resulting in a tie (where the male and female are joined for a period of time following ejaculation), the female can usually be taken home, although many breeders prefer to continue breeding the female until she rejects the male. Do not allow any other males near her while she may still be in heat, as a bitch can have puppies by two or more fathers in the same litter. This can be a disaster with a purebred bitch. After all, who will pay a good price for a registered Collie pup when two of the littermates have curly black hair like the male poodle next door?

The bitch will have her puppies after a gestation period of 58 to 62 days. As the pups grow larger inside her, she will begin to develop a ravenous appetite, and she will often eat twice the amount she ate before she was pregnant. She will need this food as well as an extra protein source, such as lean meat, cooked liver, or hard-boiled eggs. She should also receive adequate exercise. A too-fat or too-soft bitch will often have whelping trouble. She should not, however, have rough exercise, as it could cause abortion.

Whelping

Give your bitch a whelping box a week before she is due so she can get used to it. Some bitches will insist upon having their pups in a closet, in the barn, or on your bed anyway, but you can try to persuade her to use the box you provide for her. Place the box in a warm, hidden location where she will not be bothered. The sides should be high enough to prevent drafts and to keep the puppies in the nest. And they should be low enough that she can escape the pups to get exercise and take care of her own needs. Some breeders, especially those having large, heavy breeds such as the St. Bernard, like to use a whelping box with a rail to give added protection against the mother lying on the pups.

You can be quite sure when your bitch will have her pups by taking her temperature. It will drop from her normal temperature of 101° or 102°F (38.3° or 38.9° C) to 100° or 99°F (37.8° or 37.2°C). When it falls below 99°F (37.2°C),

she will whelp within 12 hours. When beginning labor, she will pant, dig, push her nest material around, and act restless. If she does not produce a puppy within 3 hours, call your veterinarian.

Most bitches will have their pups and take care of them without any assistance from you. But it does pay to keep watch on her from a distance to make sure all is well. If a pup is born and is still in the sac, break the membrane, and rub the pup dry with a clean towel; otherwise, the pup may drown in the fluid.

If the bitch goes longer than 30 to 45 minutes between pups and you feel she is not finished, take her out, and force her to take some mild exercise. If she does not produce another pup within 30 minutes after returning to the whelping box, call your veterinarian. She may have an obstruction, or her uterus may be tired and require an injection of posterior pituitary to get things started again.

After whelping she will relax and begin caring for the pups. Unless the bitch has a very large litter, she should be allowed to eat the placentas. Eating too many can cause digestive upsets and diarrhea, but eating them stimulates uterine contractions and helps to let the milk down. If it appears that she doesn't have milk, call your veterinarian immediately. If given an injection right away, she may be forced to let her milk down. Such an injection will not, however, produce milk in a bitch that does not have milk in the first place. (See "Orphan and Rejected Puppies" on the opposite page for guidance on dealing with this problem.)

The bitch will have a normal discharge from the vagina for several days or a week after whelping. This is most often bloody and dark and is nothing to worry about. If it becomes foul smelling or looks yellowish, however, call your veterinarian.

Allow your bitch free-choice dry dog food during the nursing period so she can more easily keep up with the terrific drain on her body that nursing a litter of pups causes. She should also receive a good vitamin-mineral supplement. If she appears to lose weight, adding chopped or ground meat to her dry food may help. (For information on other possible problems with whelping, see "Eclampsia" on page 316 and "Metritis" on page 325.)

Caring for Puppies

The puppies will open their eyes at about two weeks. Soon after that time, you can begin to introduce them to semisolid foods. At three weeks most puppies will lick a bit of cooked mashed liver from your fingers. At four weeks you can

usually put a shallow pan of baby cereal mixed with cooked liver and milk in with the pups. The first few times you feed them this way, they will get most of it on them instead of in them; but they will soon learn to lap up the gruel with relish. Don't make the food too thin, or they may get some of the fluid into their lungs.

At eight weeks the pups will be eating considerable amounts of food. In the weeks that follow, you can slowly switch to dry food, ideally fed free-choice. The pups also need a calcium-vitamin D supplement for proper bone growth and the prevention of rickets. If not fed free-choice, the pups should be fed four times daily at eight weeks of age, three times daily at three to four months, and two times daily until nine months.

At eight weeks the pups should have their first adult distemper-hepatitis-parvo combination vaccination and also be checked for worms. Ninety percent of puppies have some type of worms; even if the bitch was wormed, it is a good idea to have the puppies checked before they go to their new homes.

Orphan and Rejected Puppies

Sometimes it becomes necessary to raise a litter of puppies by hand. A bitch with mastitis, no milk, or a too-large litter; a rejected pup; or a bitch that dies may leave you with this tiring, but rewarding, job.

The puppies must be fed small amounts of goat milk or puppy formula (available at your veterinarian's clinic or larger pet shops) every two hours, day and night. This has to continue for at least one week. After the week you may drop to every three hours during the night, but keep with every two hours during the day. The following amounts are about right for the average pup:

- Large breeds (50 to 75 pounds, adult weight): 2 to 4 cc milk
- Medium breeds (25 to 50 pounds): 2 to 3 cc milk
- Small breeds (10 to 25 pounds): 1 to 2 cc milk

Start with these amounts, adjusting as necessary by the look and feel of the pups. After two or three days, increase the milk gradually until it is doubled in quantity. If the pups seem very hungry after feeding, you may increase the amount of milk enough to almost satisfy them. Keep increasing as the pups grow and require more food. Do not overfeed, however, as this will cause digestive upsets and possibly death.

The easiest way to feed pups their milk is with a feeding tube. This is a

flexible rubber or plastic tube that fits on the end of a syringe. Draw the milk up into the syringe and tube, where it is measured, and then slowly push it down the pup's throat. As the tube goes down, the pup will swallow it. Be sure it does swallow, or the tube could go into the lungs. When the feeding tube is down the esophagus (judge by the length of the neck), slowly inject the milk. If you do it too fast, the milk will be sprayed and cause cramps. This method requires little experience and is fast and easy. Total time required for feeding and cleaning each pup is about two minutes.

Make sure each pup urinates and defecates after each feeding. If it does not, massage the genital area with a moist cloth or cotton to stimulate action. This is the way the mother teaches her young to defecate. If it is not done, the pups may become constipated and die.

Introduce the pups to semisolid food when they are about three weeks old, as with mother-raised pups. Hand-raised pups often take readily to semisolid food a bit earlier than pups raised by their mother. By four weeks the pups will usually lap gruel made of milk mixed with good-quality canned food and a bit of baby cereal. They can be completely weaned from milk when they eat well on their own—usually by six weeks of age.

When raising very young pups by hand, you must make sure they stay warm enough. One way is to use a whelping nest heating pad, available through many kennel supply houses. Placed in the box that houses the pups, it supplies necessary supplementary heat (as their mother's body would do naturally). Without extra heat, pups often chill and soon die. Of course, you should use caution with any electrical heating device to prevent fires.

Neutering

The decision whether or not to neuter your dog is one of the major decisions that you must face in pet ownership.

Spaying

Neutering is especially a concern if you own a bitch. An unspayed bitch comes into heat every six months and is in heat for three weeks' time. During this period she may spot blood on the furniture, rugs, and floor, and her scent will attract male dogs of all descriptions from near and far. There is often fighting among these males as they vie for her attention.

Friends and relatives may offer “helpful” advice on neutering, such as “Don’t spay her until she has had a litter, or she’ll (pick one) (1) get fat, (2) be unfulfilled by missing motherhood, (3) get lazy, (4) get mean, (5) act like a male,” and so on. There are lots more “helpful” hints—as many hints as people that give them. But the truth is this: It’s easiest and safest to spay a female before she has her first heat period. Spayed at this time, the young bitch will come through the surgery with little setback and be ready to hunt, play, or work in a few weeks. Also, spayed dogs get fat for the same reason that unspayed dogs do—people feed them too much! And spayed females do not get mean because they are “unfulfilled in motherhood.” They become mean because of ill treatment or poor inborn temperament.

Spaying saves the female from the stress of raising countless puppies and possibly helps her avoid uterine infections and breast tumors. It also saves generations of unwanted puppies from being born, which would lead short, miserable lives in most cases. Animal shelters and dog pounds are full of such unfortunates, killed by the hundreds every day for lack of homes.

Spaying consists of complete removal of the uterus and ovaries. The dog is given a general anesthetic and prepared for surgery. (There are several methods of surgery, so keep in mind that the method used by your veterinarian may differ from this one in some aspects.) The hair is clipped from the belly. The line of incision is scrubbed and then painted with an antiseptic. While the dog is resting on her back, the incision is made a short distance behind the umbilical scar. The incision is carried through the skin, through the muscle, and into the abdominal cavity. The ovaries are brought up one at a time by means of a hook. Each ovary is clamped off from the ligament and tied off to prevent bleeding. Then the uterus is clamped off, just toward the vagina from where the two horns of the uterus join (the bifurcation). This is tied off securely. The uterus and ovaries are removed from the body. The peritoneum and muscle layer are sutured closed, followed by the skin incision. The dog is then moved to a recovery cage to come out from under the anesthetic.

Of course, this surgery is something you’d have your veterinarian do. But I’ve included this description because the average person does not have the slightest idea of what is involved in a spay and is often afraid to ask the veterinarian about it at the risk of appearing stupid. The spay, or ovariohysterectomy, is a major surgery, but when done on a young female in good health, there is little risk of complication.

Castration

A male dog does not present as great a problem in the home as the female. He is most often castrated because of his wandering instincts. As with the neutered female, a castrated male does not change his appearance, temperament, or working abilities. A male dog is usually castrated at about eight months to one year of age, before he starts wandering about, searching for females in heat. As castration is not abdominal surgery, it is usually less expensive than spaying the female.

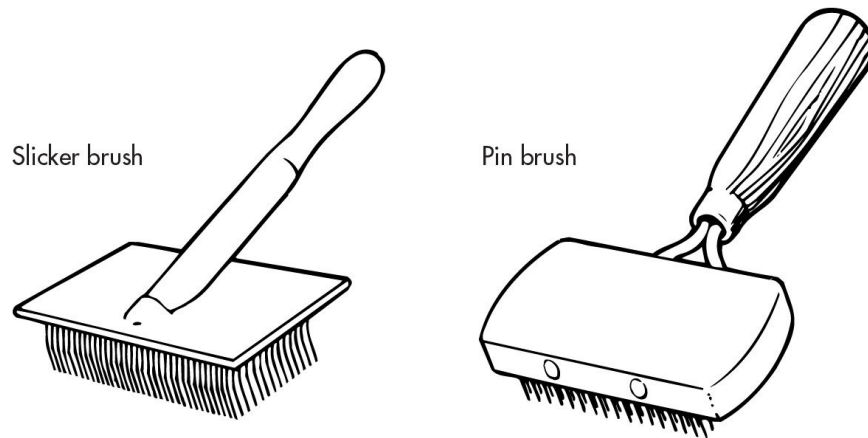
Grooming

In certain areas a dog kept in the house needs some special attention to maintain a healthy coat. As our homes are often warm and dry in the winter, many dogs shed a little (or a lot) all winter, and the coat may become dull and stiff. Even a short-haired dog needs regular grooming, not only to keep it looking good but also to keep oil in its hair and to remove dead hairs. Another benefit of regular grooming is that you'll discover any injuries or skin problems quickly, before they cause trouble.

The best way to groom short-haired dogs is with a stiff brush and chamois cloth. Be careful not to have too stiff a brush, or you may scratch a thin-skinned dog during grooming. After brushing the coat thoroughly in the direction in which the hair grows, rub the chamois a few times over it to add extra sheen to the coat. Thorough grooming of a short-haired dog, other than an occasional bath, takes only 15 minutes at most and will greatly improve the health of the dog's coat.

On a medium-coated dog, such as a Chesapeake Retriever, go over the coat first with a slicker. (This is a flat brush with bent metal pins.) A slicker is very effective for removing dead hair and stimulating the skin. Then use the regular brush and chamois.

Long-coated dogs, such as Old English Sheepdogs and Afghan Hounds, need more frequent, thorough grooming. Use a pin brush first and then a comb. (Using a slicker will tear hair out, and using the comb first will cause severe pulling if you run into mats.) On long-coated breeds, always groom from the bottom of the legs upward to be sure that the coat is mat-free all the way through, not just on the top layer.



Brushes for grooming. *A slicker brush, with bent metal pins, is useful for grooming a medium-haired dog. On long-haired dogs, use a pin brush (with straight metal pins) instead because a slicker can damage the hair.*

While grooming, watch carefully for any fleas, ticks, or red, irritated areas. Taking prompt care of such problems will bring a faster cure and prevent further trouble.

Trimming Nails

Check your dog's nails periodically. Some dogs, especially house dogs that live on carpeted floors, fail to wear down their nails naturally, and the nails can grow so long that they spiral until they stick into the pad. Such a condition is very painful and can bring permanent lameness if not taken care of quickly. Normal, well-worn nails are about even with the bottom of the pad of the foot, so that when the dog walks, the nails just brush the ground. Nails should also have a blunt appearance. When they begin to develop a hook, they are too long. When your dog walks across the linoleum or wood floor, you should not hear "click, click, click."

A pet nail trimmer is inexpensive and a handy thing to have on hand. With a little practice, almost anyone can keep his or her dog's nails trimmed and neat. When beginning, take only a little bit from each nail. There is a vein that runs down inside the nail; if the nail is clipped too short, it will bleed. Should this occur, a daub with a styptic pencil will stop the bleeding.



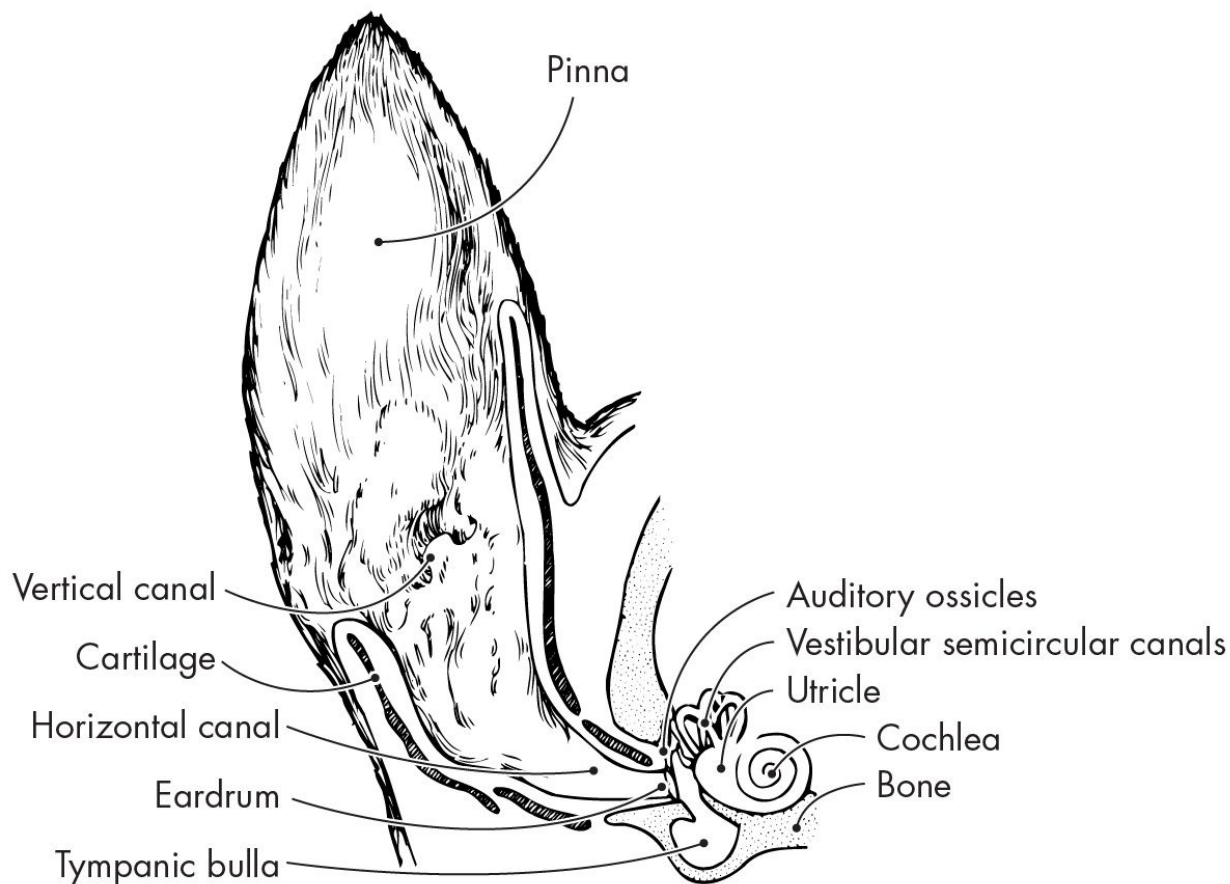
Trimming the nails. *With a little practice, you can keep your dog's nails trimmed. Just take a little bit with each snip. Avoid cutting into the quick, the fleshy portion of the claw.*

Caring for Ears

Checking your dog's ears regularly is an important part of good care. Not only do they frequently need cleaning, especially on long-eared dogs, but they are also a favorite hiding place for ticks.

If you see any brown discharge and wax, clean the ears. Using a ball of cotton moistened with alcohol or peroxide, remove the wax from as deep in the ear as your finger will reach. Using a cotton swab can be dangerous if you are inexperienced or if the dog jumps, for it can puncture the eardrum. If the wax in the ear seems very dry, putting a few drops of vegetable oil in the ear and then massaging the base well will often soften the wax. In a few hours, clean the ear as usual, and the wax will be easy to remove. The oil will also serve to soothe any itching and may smother ear mites, as well.

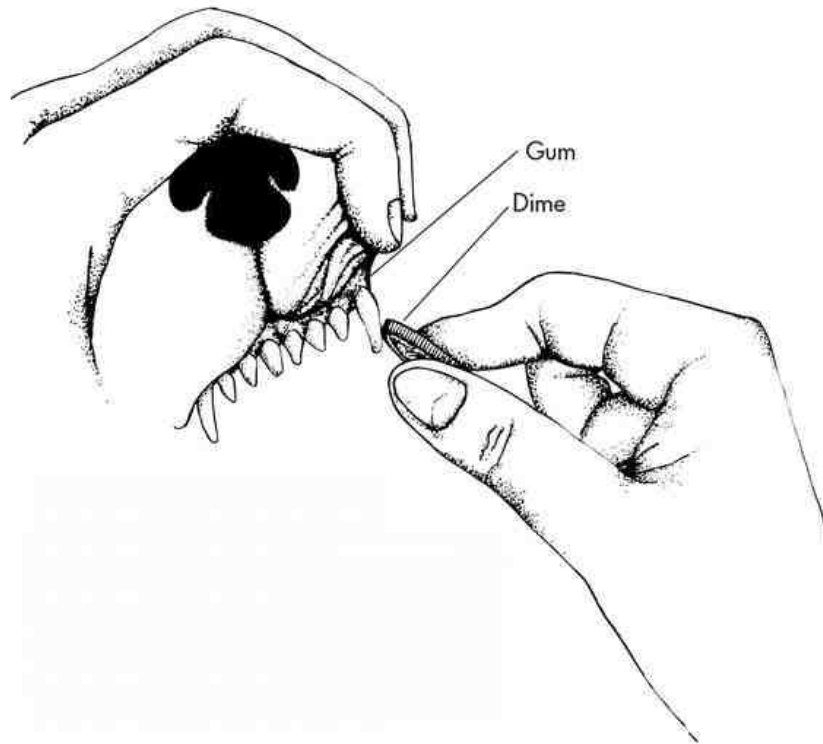
Keep a close watch on an ear that has shown a discharge. If the discharge continues or the dog shakes its head and digs at its ear, take the animal to your veterinarian as there may be an infection or ear mites present. (For more information on these problems, see "Ear Infection" on page 316.)



The ear. When cleaning the ear, use only a cotton ball and your finger. Inserting a narrow object could easily injure the delicate structures inside the ear.

Caring for Teeth

Dogs very seldom have any dental problems, unless caused by disease or old age. The main trouble with dogs' teeth is the diet they are fed. A soft diet, such as one consisting mainly of canned food or table scraps, does not keep the tartar cleaned from the teeth. As tartar builds up, it works down into the gum, often causing infection and soreness. Regularly giving a few hard dog biscuits or large bones to gnaw on will often prevent this trouble, as will keeping your dog on a diet of dry dog food.



Removing tartar. *You can remove minor tartar deposits at home; simply use the edge of a dime to scrape the tartar off the tooth. Have your veterinarian remove heavy tartar deposits.*

If tartar deposit does occur, you can often remove it yourself using a dime. Press the little rim of the coin down just beneath the edge of the gum until it is below the brownish tartar. Then scrape off the tartar by pressing quite firmly against the tooth. It usually comes away in a large flake. If the tartar deposits are heavy, have your veterinarian remove them. You should also have him examine your dog if you notice any foul odor or redness in the mouth.

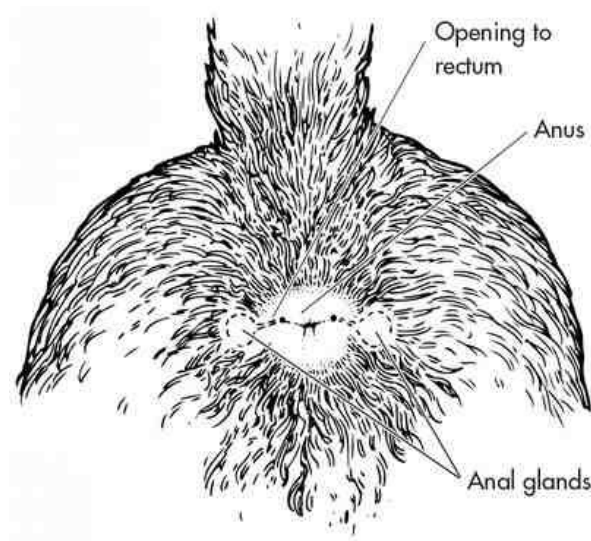
If your dog has had dental problems in the past, using a canine toothbrush and dental cleanser can help minimize further problems. Some folks feel silly brushing their dogs' teeth, but it is no more foolish than brushing your own—and your pet dog cannot do it for itself! (Wild canines “brush their teeth” by daily chewing and gnawing bones, hide with the hair on, roots, and other hard objects.)

Checking Anal Glands

Every dog naturally has two scent glands, called anal glands, one on either

side of the rectum. Most dogs should have these checked at least twice a year; smaller dogs, monthly. Your veterinarian can show you where these glands are and how to express the accumulation in them to prevent problems. Plugged anal glands can abscess and cause lameness and even hindquarter paralysis. An easy procedure is to routinely express the anal glands while bathing your dog, as any odor is then easily washed away. And there *is* a foul odor from scent material in those glands!

Expressing the anal glands is easy to learn to do. With a wad of cotton or tissue in your hand, raise the dog's tail. Place your thumb on one side of the rectum, *beyond* the location of the anal gland, and your second finger beyond the other anal gland on the other side of the rectum. Gently but firmly press in, as though you were trying to get behind the glands, and squeeze firmly. This will force any accumulation in the anal sac into the rectum and out onto your cotton. If the procedure was a success, you will notice an odor and be able to see a brownish gray liquid on your cotton or tissue. When you do this routinely, there is seldom any problem with the anal glands. (For more information on preventing problems, see "Anal Gland Impaction and Infection" on page 308.)



The anal glands. Dogs have one anal gland on each side of the rectum. Checking your dog's anal glands regularly can help prevent impaction and infection.

Caring for Aged Dogs

Older dogs often need special care to live out their lives in the best possible health. They may, for instance, need extra dental attention as their teeth wear down. Older dogs are often fed canned food, which does not remove the deposits of tartar that accumulate on the remaining teeth. These tartar deposits work into the gums, causing sore and inflamed gums, and then infection sets in. The teeth loosen, and the dog is unable to eat. Usually this infection finally claims the life of the unfortunate pet.

Check the teeth of your older dog regularly for irritation and tartar. If there is any redness, foul odor, or sign of loose teeth, take the dog to your veterinarian for treatment before the situation becomes worse. Remove tartar accumulations as they form so they don't build up into thick, irritating scales. When your aged dog's teeth receive periodic examination and treatment, there is little cost to you, and your dog will live a longer, much happier life.

Heart Trouble

Older dogs, like older people, often develop heart trouble. Overweight dogs are especially prone to this condition. Dogs need exercise and a good diet, especially as they age, to stay in good shape. There are now special diets for geriatric dogs, which tend to put on extra weight as their activity slows down.

Often the first signs of heart trouble are coughing and enlargement of the abdomen due to ascites (fluid accumulation in the abdomen). A dog with heart trouble will often have poor circulation, with fluid settling out of the blood and into the tissue. You can often control this fluid accumulation with diuretics, which stimulate the kidneys, drawing fluid out of the body. A special diet, low in salt, usually helps. Your veterinarian may recommend oral drugs to strengthen the heart; follow his instructions carefully. He can also advise you on a correct diet, which will greatly aid in maintaining the heart patient.

Urinary Incontinence

Aged dogs sometimes have trouble with urinary incontinence. This means the dog cannot control urination completely and may dribble urine unknowingly or be unable to hold the urine all night. It is most often seen in spayed, aged females.

Check with your veterinarian for advice, as most dogs can be helped with this problem. Diethylstilbestrol is often given daily, usually in oral form. The dosage is gradually reduced as the dog responds, until it is kept on a low-level

maintenance dosage. Geriatric tablets containing vitamins and minerals often help in mild cases. Making sure the dog is allowed outside often to remind it to relieve itself will also help. *Do not restrict the dog's water, because you could cause kidney problems.* If the dog dribbles urine, using “bitch in heat” panties with a sanitary napkin, changed regularly, in between trips outdoors will eliminate household problems.

Breast Tumors

Breast tumors are quite often seen on unspayed, aged females. They may or may not be malignant. Have your veterinarian check any breast lumps as soon as you find them. With some tumors injections of testosterone will result in regression of the growths. Spaying the female and removing the tumors will also help in some cases. Surgery is recommended for patients in good health, while the tumors are still small in size. Once they are large, the chances for successful surgery are greatly reduced.

Arthritis

Arthritis is a common problem with older dogs, again most commonly in heavy, obese dogs. It is important to keep your old dog from getting fat. There is a tendency toward stiffness in old age, and extra weight only intensifies the problem, making it crippling in many cases. Treatment with drugs, such as cortisone, aspirin, and phenylbutazone, will often help dogs with periodic pain. Keeping the dog in warm, dry quarters and off cement will aid in preventing stiffness. Make sure your older dog has extra bedding to protect its joints.

Often confused with arthritis, especially in older dogs, is lameness due to long toenails or ingrown nails. Because older dogs exercise less, their nails do not wear down normally, and they require periodic trimming. Otherwise, the nails will grow long, putting strain on the tendons and muscles of the leg. If allowed to grow even longer, they will curl, twist, and even grow into the pad of the toe. This causes terrific pain, and the dog will be very reluctant to move about; when it does, it will limp badly. You can very easily learn to trim the nails from your veterinarian or local dog groomer. (Also see “Trimming Nails” on page 301.)

DISEASES AND OTHER PROBLEMS

While most dogs are very healthy, it's worth your while to learn about the diseases and problems that can occur. This way, you'll know how to prevent them and how to handle them if they do happen. It's also important to know the

normal temperature of your dog so you can check for fever or other abnormalities. The normal body temperature of a dog is 101° to 102°F (38.3° to 38.9°C).

Sample Vaccination Schedule for Dogs

Use this suggested vaccination schedule as a basis for getting your dog the shots it needs. But be sure to consult your veterinarian for his specific recommendations based on your locality, the prevalence of particular diseases in your area, and the particular products he uses.

	DISEASE	WHEN TO VACCINATE
PUPPIES	Distemper, hepatitis, parvovirus (combination)	8 weeks
	Distemper, hepatitis, parvovirus, leptospirosis (combination)	12 and 16 weeks
	Rabies	3 months or older
DOGS	Distemper, hepatitis, parvovirus, and leptospirosis (where Lepto is a problem or your dog will be travelling where it may occur)	Yearly
	Rabies (booster)	Every 3 years

According to the American Veterinary Medical Association, dogs at low risk of disease exposure may not need to be given a booster yearly for most diseases. Consult with your local veterinarian to

determine the appropriate vaccination schedule for your dog. Remember, recommendations vary depending on the age, breed, and health status of the dog, the potential of the dog to be exposed to the disease, the type of vaccine, whether the dog is used for breeding, and the geographical area where the dog lives or may visit.

Anal Gland Impaction and Infection

On either side of the rectum, your dog has a small gland, which serves mainly as a scent gland. You'll notice that when strange dogs meet, they smell each other's rectum. This is to identify each other by means of the scent particular to each dog. These glands empty into the rectum during a bowel movement or at times of fright. The opening into the rectum is narrow, however, and sometimes it becomes plugged with fecal material, so the gland cannot empty. Thus, the gland fills and can become infected.

A dog with a full anal gland will drag its rear end on the ground or carpet in an attempt to squeeze this accumulation out of the gland, as it is becoming uncomfortable. People sometimes think that this means the dog has worms since itching of the rectal area is a symptom of pinworms in humans. Dogs do not have pinworms. Worming a dog with infected anal glands can be dangerous: Many times constipation goes along with the condition, and if the dog cannot get rid of the wormer, it may make the dog sick or even kill it.

Your veterinarian can show you where the anal glands are and how to express them yourself. (Also see "Checking Anal Glands" on page 304.) Doing this regularly will help reduce the chance of infection. Feeding a diet of dry dog food or giving several hard dog biscuits daily can also help prevent the condition as soft stool cannot express the glands. Feeding a diet that will give a firm stool will aid the dog in expressing the glands naturally. Exercise will usually help, as well.

Infected anal glands can cause several problems, such as lameness (due to pressure against the spine), itching, and tonsillitis, when the dog licks at infected anal glands in an attempt to relieve and soothe the area. In fact, most dogs that have tonsillitis also have infected anal glands! If your dog has recurrent trouble with the anal glands, your veterinarian can remove them.

Back Problems

Some breeds of dogs are quite prone to back problems, arising from the protrusion of a spinal disc. Long-backed breeds, such as the Dachshund and Pekingese, are prime candidates due to their conformation. For discussion of this condition, see “Intervertebral Disc Lesions” on page 322.

Breaks

The bones most often broken on dogs are the legs, perhaps because of their long length and relative vulnerability. Car accidents, traps, blows, and gunshot wounds all often result in broken limbs.

The broken leg will usually dangle in an unnatural manner. The dog is not able to place weight on it. There are extreme pain and lameness. With a compound fracture (break in skin and bone), the end of the bone will be seen sticking through a break in the skin.

Keep a dog with a broken leg quiet, and handle it gently. If the dog is in pain, slip a gauze muzzle around its muzzle and behind its ears to prevent any accidental bites. Where a human might groan or scream in pain, the dog will bite, no matter how much it loves you—it just can’t help it.

Be careful not to cause trauma to the leg by roughness. Blood vessels can be torn by the sharp end of the bone. However, do not try to splint the leg, even for temporary protection. Many incorrectly placed splints have caused more damage to a break than the dangling leg. Using care, get the dog to your veterinarian as soon as possible.

The break can be set with a plaster cast, splint, or intramedullary pin, which goes up through the bone marrow (best in many breaks). Most broken legs mend completely, with no difficulties, if you give the dog nursing at home.

Car Accidents

When a dog has been struck by a car, the first two things you should worry about are shock and internal bleeding. A broken leg or gash is a minor worry compared with these problems. Check the dog’s color (see if the gums are pink or pale white, indicating shock), and get the animal to your veterinarian immediately. The best way to move a small-or medium-sized dog is to first muzzle it and then grasp the skin behind the neck and on the back, gently lifting the dog to a blanket or rug. Grabbing the dog around the waist or chest can stick a broken rib through a lung.

Do not try to give the dog any food or water. An injured dog is usually in shock and is not in any condition to take food or water, even if panting. Severe vomiting can result, which could start internal bleeding. And should the dog need immediate surgery, food or water in the stomach could make the surgery dangerous. Let me repeat here: Take the dog to a veterinarian immediately. Often an injection of a clotting drug, a stimulant, or immediate surgery can save a dog that would die otherwise.

After the first 24 hours, there is another danger period that can last for about four days. It is in this time that occasionally a dog is lost from a blood clot. The dog may be acting fine—it is active and eating—then suddenly die. (This also happens in humans.) You can guard against this somewhat by restricting the exercise the dog takes and feeding small amounts of easily digested foods several times daily. Your veterinarian may also prescribe drugs, taken orally, in some cases. There is *no* “sure cure” preventive.

Coccidiosis

Coccidiosis is caused by coccidia, parasitic protozoa (one-celled animals). It most often affects puppies, although adult dogs are often carriers and can infect younger dogs or become affected themselves when under stress. Coccidiosis causes diarrhea, coughing, and dehydration. There are often secondary infections, such as pneumonia and distemper. Damp, unsanitary conditions and overcrowding help to spread coccidiosis, as it is passed by contact with fecal material containing eggs or oocysts. A fecal examination is necessary both in diagnosing and in treating coccidiosis as many other things can cause diarrhea, especially in puppies.

Combination intestinal sulfas will usually work in treating coccidiosis. I have also had good success using cottage cheese. This seems to change the pH of the intestine, killing the coccidia. I have had pups that did not respond to sulfa clear up immediately when fed cottage cheese as a sole diet. I know it sounds like a quack remedy—I chuckled the first time a veterinarian friend told me about it—but it does usually work and causes no side effects, as sulfa drugs sometimes do in cases of dehydration.

Pups with diarrhea also benefit from receiving electrolytes in their water, until the stool is firm again, to combat dehydration that can kill them.

Conjunctivitis

Conjunctivitis is inflammation of the conjunctival sac next to the eye. The area will be red and swollen and appear bloodshot. The eye may run or stick shut. There may be pain or itching. Often there is a bacterial infection following the initial irritation. Conjunctivitis commonly starts with a foreign body, such as dust, pollen, or weed seeds, blowing into the eyes. A good way to have eye problems with a dog is to let it hang its head out the car window while you are driving. The speed of the moving car adds great force to any grit or dust blowing in the wind, driving it into the eye or conjunctiva. If left untreated, the condition can get bad enough to cause blindness.

Often, treatment consists of flushing the eye with a mild solution and applying an ointment containing an antibiotic or a sulfa and an antiseptic to ease the pain and itching. Rubbing the eye only increases the injury. Also, tape the dog's dewclaws to its legs to prevent the dog from scratching its eye. If it paws at the infected eye, it can hook the eye or the rim of it and cause serious damage.

Conjunctivitis is usually a simple condition but can be a symptom of certain diseases, such as distemper; so it is best to take the dog to your veterinarian, just to be safe.



Giving eye medication. Steady the head, and pull down the bottom eyelid to make a pouch. Resting your other hand on the head, drop the medicine in the pouch.

Cuts, Bites, and Other Wounds

Dogs suffer the same types of injuries humans do, including cuts, bites, and scrapes, and the same types of basic first aid work for both. First, stop the bleeding by applying pressure to the wound with a gauze pack. Call your veterinarian at once unless the injury is minor. If it is minor, you can treat the wound yourself. Clip the hair from the edges, and clean the wound gently with warm, soapy water. Do not use Lysol or any such strong disinfectant, as you may be killing more cells than you are cleaning. Also, do not bandage the wound. Bandages retard healing and often cause infection by providing a warm, protected, moist area for bacteria to breed. And do not put ointment onto a wound, as it usually attracts dirt, hair, and other irritants. A liquid antiseptic, such as Betadine, is much better because it dries quickly.

Few wounds on a dog actually require sutures. People usually prefer to have a wound sewn closed, often just so they do not have to look at the ugliness. But wounds usually heal quicker and without scarring if they remain open, healing from the inside. Keep an open mind when asking your veterinarian if he thinks a wound should be sutured. Some should be, of course, but certainly not all of them.

If your dog gets bitten, use special caution as it is always possible that the animal that inflicted the bite was rabid. If you know what animal bit your dog, cage the animal, if wild or stray, taking extreme care *not* to get bitten yourself. If the animal belongs to a neighbor, make *sure* that it has been vaccinated against rabies. Also, ask your neighbor to watch it for two weeks and let you know if it should get sick or die. (Generally, after two weeks' isolation, an aggressive animal will show definite symptoms of rabies or die.) When washing your dog's bite, use rubber gloves and plenty of soapy water to thoroughly flush out all bacteria, possible viruses (such as rabies), and dirt. Do *not* get in contact with saliva from the bite wound.

If you suspect your dog may have been bitten by a rabid animal and your dog is current with its rabies vaccination, revaccinate it immediately, and confine it, using extreme care, for 90 days. Should your dog be bitten by a stray or wild animal, the safest route is to have the offending animal destroyed immediately, taking care to save the entire head, and have the brain tested for rabies. If your dog is unfortunate enough to be bitten by a suspected rabid animal and is not current in its rabies vaccination, the animal should either be destroyed or be completely confined for six months and be vaccinated against rabies four weeks

before being freed from total confinement. During the confinement *extreme* care must be taken to prevent possible human infection as rabies may be spread before the dog shows any sign of having the disease. (For more information on this disease, see “Rabies” on page 327.)

Cystitis

Cystitis is an inflammation of the bladder. It is quite common in dogs but not always diagnosed, as many times there is just a mild inflammation that causes no visible problems. Cystitis is most frequently caused by bacterial infections and calculi (stones).

A dog with cystitis often urinates frequently, vomits, dribbles urine unconsciously, and acts restless. There may be a fever. Lifting the dog may bring yelps of pain. There is often blood in the urine. Where the cystitis is caused by a stone (a bit more common in males), there may be a complete blockage in the urethra. When the dog is not able to urinate, it absorbs the wastes and soon becomes toxic. Within a short time, the dog is in real (and possibly irreversible) trouble.

As you can see, prompt discovery of the problem and quick diagnosis and treatment are very important. With an infection in the bladder, antibiotics usually bring about prompt relief. Surgery is generally the best treatment for a dog with stones, as they are usually quite large when causing the dog problems and they are nearly impossible to dissolve. Following surgery your veterinarian may put your dog on a special diet or advise putting a tablespoonful of vinegar in the drinking water to acidify the urine, which often helps prevent the formation of calculi in the future.

Diabetes

Diabetes is a disease that strikes dogs as well as people. It most often occurs in fat dogs over four years of age. The first signs are an increased thirst and frequent urination. Then there is a dramatic weight loss, quickly getting to the point of emaciation. The dog begins to act weak. In severe cases there is uncontrollable vomiting.

When caught fairly early and when there are no complicating factors, diabetes can be controlled quite easily if you are willing to spend a little extra time with your dog daily. Diet and exercise alone can control some mild cases. Reduce the carbohydrate intake severely, and give the dog foods like lean meat, boiled eggs,

and boiled fish. Other cases respond well to oral medication. Still other dogs need insulin injections daily, given before regulated meals. If you give the injections religiously, you can maintain a diabetic dog quite well with the insulin. The needle used is very small, and the injection is practically painless. The dog soon associates the injection with eating and often begs for the injection so it can eat! Treatment and maintenance are inexpensive and can do much to prolong the life and health of a diabetic dog. Preventative measures include maintaining an active lifestyle in your dog, including regular exercise and keeping your dog within a good weight (not overweight!).

Distemper

Distemper is a disease of the central nervous system caused by a virus. It is usually severe, and there is no certain treatment. Distemper is very contagious among all canines, not only dogs. The common signs are mattered, running eyes, a plugged, crusty nose, diarrhea, and vomiting. Chorea—twitching of a leg, one side of the body, or tail—usually develops when the disease is nearing its worst.

Distemper is often confused with rabies, also caused by a virus. There is no similarity between them. You *cannot* get distemper if bitten by a dog ill with distemper; you *will* get rabies after being bitten by a rabid dog unless you get treatment. A dog sick with distemper may be crabby, but biting is not a symptom of distemper.

Distemper may occur year-round, and it is so highly contagious that it does not even need close contact to spread. You can bring it home on your clothes, or a dog with distemper can just pass through your yard and spread the virus to your unvaccinated dog.

Treatment, once the disease has started, is often frustrating for both the owner and the veterinarian. The sick dog may look better and have signs of recovery one day but then get worse the next. Antibiotics will not cure the disease, as it is a virus, but they can prevent secondary infections, such as pneumonia, in a very stressed dog. Injections of antiserum or globulin can help some dogs.

Good nursing is very important in treating this disease. Without it most dogs will die. A dog sick with distemper will not want to eat, so it must be coaxed to eat, force-fed, or tube-fed. Also, keep the dog warm and clean. Clean the eyes and nose several times daily, treat the eyes with ophthalmic ointment, and moisten the nose with mild petroleum jelly. A dog having convulsions must sometimes be given drugs to combat them so that it does not become overtired.

As with many other diseases, distemper is easier to prevent than treat. A puppy born from a dam that is immune to distemper through vaccination will have a certain level of immunity from its dam's milk. This immunity wears off at about eight weeks, however, so the pup must be vaccinated to remain protected. This first vaccination is usually done at eight weeks of age. It is often given in a combination shot containing distemper, hepatitis, and parvovirus. Leptospirosis used to be included in this first injection. But as it is quite rare now and because it is believed that early lepto vaccination can tie up a pup's immune system, making it less responsive to the distemper, hepatitis, and parvo portion of the vaccination, it is now usually deleted until a booster vaccination, given several weeks later.

If necessary, puppies may be vaccinated earlier, at three weeks of age, with a "puppy shot," or temporary vaccination. Puppies not receiving their vaccinated dam's milk or those under stress or in a high distemper area often do well with this added protection. As opposed to the older belief, there is not a "permanent" distemper vaccination. To keep active immunity, the dog should receive a yearly booster, once it has received its initial series of vaccinations.

Ear Infection

Dogs, especially long-eared dogs, are prone to several types of ear infections. The most common is due to ear mites. Ear mites burrow into the ear, creating places for bacteria and fungi to enter. The moist, dirty ear that often accompanies the mites makes a perfect breeding place for these infections. Bacterial and fungal infections can also get a start from irritation due to dirt and wax buildup, water or shampoo in the ear, foreign bodies, or injuries.

The dog will usually scratch at its ears with its hind feet and rub its head on the floor or against a chair. It may also shake its head, hold it to one side, and appear uncomfortable. When you look in the ear, it will often appear red and swollen, have a brownish, foul-smelling accumulation inside, and feel hot.

Mild cases, caused by accumulations of wax and dirt, may be taken care of at home. Dip a ball of cotton into alcohol or peroxide, and gently but firmly swab out the ear by using your finger and the cotton ball. Repeat the next day. There should be a great improvement, but if not, take your dog to the veterinarian. Fungal and bacterial infections do not respond to simple treatment and can get worse if not treated properly, possibly damaging your dog's hearing. Any extreme itching or aggravation should be taken care of at once before other problems develop, such as a hematoma (a large, blood-filled bruise of the ear,

usually caused by flopping the ears back and forth, due to irritation).

Ear mites are treated like wax and dirt accumulations, with a cotton ball dipped in alcohol or peroxide, along with adding a few drops of vegetable oil to the ear every day. (The oil smothers the mites and soothes the irritation). If there is not much improvement after two days, call your veterinarian. It is sometimes necessary to use an ear mite remedy containing an insecticide, such as rotenone or pyrethrin.

Eclampsia

Caused by a low blood calcium level, this condition in many respects resembles one called milk fever in cows and goats. It can occur before whelping, but it is generally seen in bitches after giving birth, when the puppies are a few days old. It is often a small female with a fairly large litter that is affected. She will begin to pant, walk with a stiff gait, stumble, fall, and be unable to rise. She may have convulsions later on, stiffening out rigidly. If not treated, she will likely die.

Take the bitch to your veterinarian, who will give her an intravenous injection of calcium. Most females will instantly respond and rapidly become normal in their actions. It is possible that she may need a repeat injection, sometimes in a day or so, so be watchful for the first signs of trouble. When the litter is large and several weeks old, it is best to wean some or all of them at once. If they are allowed to continue nursing, the condition will recur.

Epilepsy

Epilepsy is quite common in dogs, especially in some breeds like Cocker Spaniels, Miniature and Toy Poodles, German Shepherds, and some of the smaller house dogs. This is not to say many of these dogs have epilepsy but that it is more common in these breeds.

A dog with epilepsy will have recurrent “fits,” or convulsions. The dog shakes, stiffens out, falls, and jerks its legs; its head often tips backward. Often the dog will lose control and empty the bowel and bladder. After the seizure the dog usually recovers and acts dazed. Excitement may trigger these seizures. They may occur only once a year or several times daily in severe cases. They don’t last long each time they occur.

There are several oral drugs that can be given when epilepsy has been

diagnosed. Primidone and Dilantin, which are used with human epilepsy patients, will usually keep the dog from having seizures if given regularly. After a few months of treatment, you can gradually cut down the dosage until you reach a level that will just prevent the seizures. (One German Shepherd in our clinic required Dilantin only during periods where thunderstorms threatened, as these—and nothing else—set off his seizures!)

Do not use dogs with epilepsy for breeding, because it has been proved to be a genetic trait in canines.

Fleas

When a dog parasite is mentioned, everyone immediately thinks of fleas. But as long as you take a few preventive measures, fleas usually aren't much of a problem for most dogs. However, there are some dogs that develop an allergy to fleas. An allergic dog will break out with a rash, scratch, and be miserable if even one flea takes up residence in its coat.

Fleas are little, quick-moving, flat, brown insects that suck blood and annoy dogs and people. And dog fleas will bite people! Sandy, dry areas are favorite spots for fleas, and there are geographic locations where fleas are very prevalent. Fleas do not spend their life on the dog but hop off and on. This is why, when treating the dog with a flea powder or spray, you should also treat its bedding, rug, or sleeping area—even the house and carpet if the fleas are numerous.

An herbal flea deterrent, such as a pennyroyal mixture, either powdered on the dog or in a flea collar, often works when fleas are not numerous on the dog. If this is not successful, you can use a powder or spray containing rotenone, which is generally safe and effective. Dust or spray weekly for three weeks for surest results. Even more effective, but slightly more work, is using a shampoo containing rotenone. There are topical flea killers which also work on mites and ticks. These, such as Frontline, available through your veterinarian, are applied in a strip down the back, on the skin. They are then absorbed into the body and bloodstream making the dog's blood toxic to biting insects. Your dog may get fleas on him but when they bite, they will then die. I only recommend these when fleas are a problem on your dog.

If your home gets infested with fleas (which can happen even with no animals in the house), it is best to call an exterminator. Most flea sprays on the market will kill fleas, but the exterminator will be experienced in knowing the favorite hiding places and will know how to get maximum kill with a minimum of spray.

It is a good idea to tell your exterminator that you are concerned about using undue toxic chemicals in your home, as there are certainly alternatives!

Flies

Biting flies are often a problem for an outdoor dog. Dogs with erect ears, in particular, are often attacked so badly on the ear tips that they bleed and scab over. These scabs in turn attract more flies. A daily application of a wipe-on insect repellent, such as that used for horses, will prevent this. If the dog has already been bitten, daubing scarlet oil on the raw spots will help heal them and repel flies at the same time.

Besides biting, flies also cause trouble by laying eggs on “woolly” areas of the dog, especially if they are soiled by diarrhea or blood from an injury. The rear parts of such breeds as Cockapoos, Collies, Cocker Spaniels, and Old English Sheepdogs are common sites of fly strikes. Soon after the eggs are laid, they hatch out into maggots, which not only eat the filth but quickly begin to destroy living flesh, as well.

Immediately examine any dark, foul-smelling areas, and wash them thoroughly if there are no maggots present. Clipping the area will also be a help. If you do see or suspect maggots, call your veterinarian at once (within a day or two) as the maggots may eat their way into the dog’s abdominal cavity. Common treatments include clipping all the hair off in the area, washing the maggots out, flushing the open wound with an insecticidal antiseptic, and giving antibiotics to prevent infection.

Fungal Infection

Many types of fungal infections can attack dogs, but perhaps the most common is the fungus that causes “hot spots.” Hot spots often appear very suddenly during the summer and fall. They are bald, angry red spots that have a moist, whitish look in the center. They either itch intensely or are very tender, causing pain if touched lightly. They usually occur on the neck and back, and they spread outward, generally keeping a roundish shape.

Have your veterinarian examine your dog when you suspect this problem. He can prescribe an antifungal, which quickly clears up the trouble. You’ll need to clip the hair away from the entire red area so you can be sure the medication will reach the skin and to keep the hair from causing further irritation.

Other types of fungal infections affect the skin in less dramatic ways, but they still produce red, scabby, itchy lesions that irritate the dog greatly. When you suspect any fungal infection, take your dog to the veterinarian to have the problem correctly diagnosed and begin a treatment program. Many times the only treatment needed is a medicated bath. Other times you'll need to use local treatments after the bath or give antibiotics when a secondary bacterial infection is also present.

Gunshot Wounds

For some reason, dogs receive more gunshot wounds than any other animal. Perhaps it is because they do more “roaming” into other people’s yards and are “chased off” at gunpoint—a very good reason to be *sure* your dog stays home, under supervision. Also, during hunting season some hunters will shoot at anything, be it a sparrow or dog. Knowing where your dog is at all times will protect it from such wounds.

Being shot with birdshot (many small pellets) is often more dangerous than being hit with a single rifle bullet. This is because rifle wounds are more noticeable and tend to get treated right away, while pellets from birdshot often leave very little outward sign when they penetrate the body.

If you suspect your dog has been shot, examine it carefully. Really pick and feel. With birdshot you will want to know where the majority of the shot struck. With a rifle wound, look for an entrance wound and exit wound. The exit wound may not be in line with the entrance wound, as the bullet often deflects off a bone.

Keep the injured dog warm and calm. Stop any bleeding that is present using a pressure pack. Then call your veterinarian at once.

Hematomas

Dogs sometimes get bruises, as people do. However, a dog’s skin is looser, allowing more bleeding under the skin (a *bruise* is bleeding under the skin). These bruises fill the area with blood, making a lump called a hematoma. Hematomas are often found on ears, due to flopping the ears in attempts to get rid of itching ear mites or dirt accumulation. They’re also commonly found on the neck from collar injuries, such as those that occur when a dog lunges on a chain, or on the elbow of a large dog that lies on concrete or other hard surfaces.

With an ear hematoma, the layers of skin pull away from the ear cartilage, and this can cause severe thickening and deforming of the ear flap. Ear hematomas can sometimes be eliminated without surgery if caught early, but often surgery is required to save the normal appearance of the ear. With other kinds of hematomas, it's best to leave them alone unless they are large or in a place where they might cause trouble, such as near the mouth or genitals; then take your dog to your veterinarian for an examination and treatment suggestions.

Many animals have bled to death when owners have lanced a hematoma, thinking they were lancing an abscess. But once relieved of the pressure that stopped the bleeding under the skin, the animal continues to bleed, sometimes so much that it is fatal. Never lance any bump before drawing some of the fluid out of it with a syringe and 18-gauge needle. The fluid in an abscess is yellowish or whitish. If the fluid is dark in color, suspect a hematoma: Do *not* lance it.

Hepatitis

Hepatitis is caused by a virus and very contagious, often spread by the urine of infected dogs. Its symptoms resemble those of distemper. There is usually a fever, inflamed eyes, vomiting, no appetite, and diarrhea. Hemorrhages and hematomas sometimes occur.

If you suspect your dog has hepatitis, call your veterinarian. Treatment for hepatitis consists of blood transfusions, intravenous fluids, and broad-spectrum antibiotics to prevent secondary infections. Since distemper often follows or accompanies hepatitis, it is usually a good idea to give globulin to prevent it, especially in a unvaccinated dog or a dog with a distemper vaccination that is not current.

A dog with hepatitis may also develop white eyes, which can cause temporary blindness. This usually follows the acute period of the disease, in which the dog is the sickest. If not treated, it will usually disappear in a week or two. This condition is called blue eye. The same condition can follow vaccination. It, as well, usually clears without treatment. Do not put cortisone ointments into the eye or give injections of cortisone, as this sometimes causes scar tissue to form, making the eye permanently blind.

As with many canine diseases, vaccination is much more satisfactory than treatment—and cheaper, as well. Most times hepatitis (canine adenovirus 1 [CAV 1] and related CAV 2, one of the causes of canine tracheobronchitis, or kennel cough) is included in the distemper vaccination combination; ask your

veterinarian.

Hip Dysplasia

Hip dysplasia is a problem of the hips. It is often not noticeable without an X ray, except when the dog is severely affected, often in later years. It is a hereditary condition, and as with most hereditary problems, it is found most often in purebred dogs of certain breeds. German Shepherds, Old English Sheepdogs, and St. Bernards are a few of the breeds with a high incidence of hip dysplasia. Inbreeding or linebreeding on lines with even a scattered incidence of the condition can produce a litter with several pups affected with various grades of hip dysplasia, and the rest of the pups can be carriers of the condition.

It is recommended that every bitch and male used for breeding be X-rayed at 24 months of age—before being bred. An earlier X ray can provide false readings. It is not enough to say that the animal appears sound. These X rays may be taken at your local veterinarian's office or a state veterinary school. They can then be sent into the Orthopedic Foundation for Animals (OFA), where the dog is issued an OFA number, provided it is found sound. If all breeding animals were examined in this way, hip dysplasia would gradually be weeded out. But this is a utopian dream as it is nearly impossible to get all breeders (or all people, for that matter) to cooperate.

Your veterinarian can treat hip dysplasia once it is diagnosed. Dogs with this condition are often maintained for years on periodic doses of buffered aspirin and anti-inflammatory drugs such as phenylbutazolidin, combined with a strict diet to limit body fat, as the added weight puts extra stress on the hip joint.

When the pain and anti-inflammatory drugs no longer seem to be working, surgery is recommended in many cases. Complete removal of the femoral head sounds very radical, but in most cases the dog continues to walk nearly as well as it did before the surgery.

Intervertebral Disc Lesions

Certain breeds, usually long-backed breeds, are quite prone to back problems arising from protrusion of a spinal disc, which puts pressure on the spinal cord. It can arise from an injury and is hereditary in a certain sense, as Cockers, Pekingese, and most often Dachshunds—all having long backs—are affected.

Jumping on and off furniture, climbing up and down stairs, and leaping in and

out of cars often cause sudden, acute pain and sometimes paralysis of the hind legs. Sometimes you'll first notice severe lameness and an unwillingness to stand or to move about. The problem quickly progresses. In a few days, the dog will often assume a froglike position, with the hind legs out behind the body.

Surgery is often used, but as with humans, not all back surgery is successful. Very often, though, these back injuries can be healed to the point that the dog can walk and lead a normal life with the help of drugs, good nursing, and physical therapy.

Leptospirosis

Leptospirosis is caused by a bacterium and can be passed to humans. There are many locales where there is a high incidence of this disease, higher than most people realize. It can be passed through dogs, rats, or wildlife, depending on the type of lepto organism considered.

The first signs may be fever, diarrhea, increased urination, bloody or discolored urine, vomiting, and weakness. There is often permanent kidney damage, even if the sick dog survives the disease. The mucous membranes in the mouth may become inflamed and sore and later slough off.

If you suspect your dog has leptospirosis, call your veterinarian at once, and use good personal hygiene when handling the dog. Antibiotics, such as streptomycin or tetracycline, are often used for treatment as well as intravenous electrolyte therapy to combat dehydration and help stressed kidneys.

Prevention consists of vaccination and of keeping your dog away from places rats are known or suspected to frequent, for rats are often the carriers of this disease, spreading it through contamination with their urine. Cattle and other dogs may also spread the disease, so it's important to control your dog to keep it away from potential carriers.

Lice

Lice are small, slow-moving, grayish parasites that occasionally infest dogs, usually outdoor farm dogs. They are most often seen on medium-or long-coated breeds as the long hair enables them to hide from the light. When lice are bothering a dog, the animal will usually begin scratching excessively. On parting the hair, you will see the tiny lice and the little specks that are their eggs. A severe infestation of lice can kill a dog through severe anemia.

When you discover lice on your dog, clip the entire dog short. With no hair to shelter and protect the lice, it's usually easy to take care of these pests, often with a medicated shampoo. Repeat treatment every week for three weeks to kill any new lice that have hatched out. Remove the doghouse bedding and burn it, and dust the dog's bed with a safe louse powder.

Lyme Disease

Lyme disease is becoming more and more of a problem in dogs as well as in people. Most frequently spread by deer ticks, Lyme disease has been found in dogs in nearly every state of the United States as well as in Canada, Europe, and Australia.

The symptoms of Lyme disease can be vague, but a fever, lameness, weakness, and swollen lymph glands are commonly seen. Diagnosis is difficult as even blood tests on humans are often incorrect and misleading. Most veterinarians will test a dog suspected of having Lyme disease, but they treat depending on the symptoms, especially in high-Lyme areas. Treatment usually consists of long-term therapy with antibiotics, such as penicillin, tetracycline, or doxycycline.

To minimize the chance of Lyme disease, remove all ticks very promptly (daily at least), taking special care not to handle or "squash" the tick with your bare fingers. Dispose of any ticks by burning; then wash your hands well. A vaccination is now available for dogs and should be given in any area where Lyme disease has been diagnosed. Consult with your veterinarian.

Mange

Mange is quite common in dogs. Perhaps this is because they have such close contact with people! You see, people can often carry the mange mites on their skin and don't realize it because the mites don't bother people. Once, a vet school class took skin scrapings from the whole class—and over 90 percent were found to have mange mites! And this was before they had any actual contact with animals. (They also weren't a dirty, scruffy bunch!)

These mites are so tiny you cannot see them with the naked eye. They burrow into the skin, causing intense itching. A mange infestation often follows a period of severe stress. Malnutrition, followed by poor skin health, often lets the mites get a head start and cause problems for the dog. Any skin eruptions, especially those about the face, legs, and belly, should be regarded as suspicious. Pustules,

scabby areas, thickened grayish areas, and baldness are all symptoms of mange.

Mange is very contagious to other dogs, so it is best to isolate the suspect until your veterinarian makes a proper diagnosis. Do not just let a neighbor tell you that your dog has mange, as there are several other skin troubles that look like mange. (See “Fungal Infection” on page 319 for more information on other skin problems.) The only way to make a correct diagnosis is by examining a skin scraping under a microscope.

A dog affected by mange should be clipped entirely. This is often hard to agree to, especially if your dog has a long, beautiful coat, but it is the only way to stop the mange at once. You can't treat spots for mange if you can't find them, and you can't treat through hair and do a decent job. Many times there are mites present in areas that do not yet show lesions.

Using medicated shampoos, along with ointments containing rotenone, lime-sulfur, or pyrethrin, will usually effect a cure. Many times you must repeat treatment two or three times, at weekly intervals, even if the skin seems much better. There are several newer treatments, including dips, such as Amitraz and Paramite and topical solutions applied once a month, such as Selamectrin, that offer quite reliable results. Ivermectin has also been used on some dogs with good results. Always consult your veterinarian, as some breeds of dogs require different treatments as some have undesirable side effects from some of these drugs.

Metritis

Metritis is inflammation of the uterus. There are two types: acute and chronic. With acute metritis the bitch is very sick. She has often retained a placenta or fetus and absorbed the foul uterine discharge, making her extremely toxic. She will vomit, have diarrhea, run a high fever, and act listless. If your bitch has recently whelped and acts dull or has abnormal uterine discharges with a foul odor, have your veterinarian examine her at once. It could save her life.

With chronic metritis there is often a low-grade infection. The bitch can often go a long time with a mild infection without acting sick. Sometimes the only sign of trouble is breeding difficulties. The bitch may be bred once or several times and not conceive. If she does conceive, the puppies may either be stillborn or die shortly after birth. Although antibiotic therapy can be used, the best course of action in many cases is to remove the uterus. Sometimes the low-grade infection can “blow up” into an acute attack, and the bitch will be lost. It is very

hard to get rid of chronic metritis without removing the uterus. With a very valuable breeding female, your veterinarian may try infusing the uterus periodically with antibiotics and using injections of prostaglandin to help push any foul matter out of the uterus.

Parvovirus

Parvoviral infection, often referred to just as parvo, is a relatively “new” disease, first recognized only in the late 1970s. There are two different forms of the disease: myocarditis, causing sudden death, often in young puppies, and enteritis, causing diarrhea, often severe and streaked with fresh blood. Parvovirus is very contagious, chiefly through exposure and ingestion of feces-contaminated food or water. Death is not always the result, but it is common with young, unprotected puppies.

If you suspect your dog has parvo, call your veterinarian at once. Treatment often consists of intravenous fluids and antibiotics to prevent or treat bacterial infections that accompany or follow the parvovirus infection. You’ll also want to isolate the dog promptly and give careful attention to cleanliness.

Prevention consists chiefly of vaccination, beginning at a young age, with revaccination yearly.

Porcupine Quills

Sooner or later, nearly every farm, ranch, or hunting dog runs into a porcupine (provided, of course, that it lives in an area where there are porcupines). These slow-moving animals prefer to just go about their business; but when harassed by a dog, they can quickly slap a lot of quills into the dog’s nose.

Many dogs have the sense to quit after they’ve gotten four or five quills stuck into their noses. But once in a while, a dog really gets mad and tries to tear the porcupine up. The usual result is a dog with a mouth, nose, and throat full of quills. When this happens, take the dog to the veterinarian for quill pulling. Any attempt to remove the quills at home will be futile.

Porcupine quills have tiny barbs that, when imbedded in the flesh, make them hard to pull. These same barbs cause the quill to work deeper and deeper in time. If the quills are grasped firmly and pulled very slowly, usually with forceps or pliers, the barbs will release. But if yanked, which is not uncommon if the dog is struggling, the quills will snap off, leaving a large amount of quill beneath the

skin. These quills often become infected, causing swelling and pain. There are also tiny quills that get into the skin when the dog bites at the porcupine. These are little bigger than a hair, but if left in or broken off, they can cause as much trouble as the 3-inch-long quills. This is why it is usually best to take the dog to your veterinarian, where it will receive a general anesthetic so it can be relaxed while the painful operation of quill pulling is done.

Once in a while, a dog gets into quills so badly that it goes into shock. In this case it must be treated at once, or it will die. If the dog seems suddenly shaky, looks pale, and acts strangely, get it to your veterinarian immediately, while keeping it quiet and calm. Do *not* attempt to feed or water a dog suspected of being in shock. This could kill it.

Rabies

Rabies is caused by a virus that is very highly concentrated in the saliva of infected animals. It used to be called hydrophobia (fear of water) because the paralysis of the throat made it hard or impossible for a rabid dog to drink.

There are two types of rabies: dumb rabies (the dog just sits, with its mouth often hanging open, and has a peculiar look in its eyes, later frequently howling strangely) and furious rabies (the dog hallucinates, snaps at imaginary objects, wanders, and becomes increasingly irritable). Often the dumb rabies is the most dangerous as many people pry into the mouth, thinking the dog has something stuck in its throat or mouth. Getting the dog's saliva into a small cut or scratch can give you rabies, so *never* stick your hand into such an animal's mouth.

Not all rabid dogs attack people and have frothy mouths. A rabid dog often has a personality change. A timid, fearful dog will get brave and friendly. A normally friendly dog will get surly and timid. A rabid dog will often leave home and wander until exhausted.

Skunks, raccoons, foxes, and bats are frequently carriers of rabies, so never try to pick up or catch a "friendly" wild animal. Keep your dog away from such animals, as well. Rabid skunks frequently bite, not spray.

If your dog is bitten by a stray or wild animal, carefully cage or kill the suspect, leaving the entire head intact. Taking great care not to become infected yourself, transport the suspect to your veterinarian for rabies testing. Immediately place your dog in strict quarantine while you are waiting for the test results. If the test proves positive (the animal that bit your dog was rabid) and your dog was not vaccinated against rabies, it is recommended that your dog be

destroyed at once. If you cannot bring yourself to do this, it must be kept in *strict* isolation for six months—which is very hard to do, especially if there are children at home—and vaccinated against rabies a month before release.

If your vaccinated dog was bitten by a known rabid animal, it is recommended that you revaccinate it and then place it in strict quarantine for 90 days.

It is recommended that any dog bitten by an unknown wild or domestic animal be regarded as having been bitten by a rabid animal and treated as such.

As you can see, vaccinating your dog against rabies and keeping its vaccination schedule current protect not only your animal but your family, as well.

Snakebites

Many dogs are bitten by poisonous snakes every year. However, don't assume that because your dog was bitten by a snake, it will certainly die. First of all, many snake bites are by nonpoisonous snakes, such as the bull snake, garter snake, and king snake, all of which bite as part of their aggressive display, meant to frighten away attackers.

Poisonous snakes in the United States encompass the pit viper family, including rattlesnakes, copperheads, and water moccasins, whose bite causes pain and swelling and will cause the area to become necrotic and slough off in time. Another poisonous snake is the coral snake, found chiefly in the southeastern United States. It is known by the old rhyme "Red touch yellow, kill the fellow," referring to the red, yellow, and black banding of its coloring. The coral snake's shortfanged bite paralyzes the victim's respiratory center, frequently causing death in a short time.

If your dog is bitten and you see the snake but are unsure if it is poisonous, kill the snake, and take it and your dog to the veterinarian at once. Do *not* get bitten yourself; "dead" snakes are perfectly capable of inflicting poison via a bite, even when the head is off the body.

It is important to know that not all bites by poisonous snakes result in injection of poison. It has been proved that over 50 percent of all bites by venomous snakes do *not* contain poison, for one reason or another. But if your dog is bitten, assume that the bite contained poison, and get your dog to a veterinarian at once. *Do not attempt first aid.* Just keep the dog calm and quiet. Carry it, if you are able, to your car, and drive immediately to the nearest

veterinarian.

Treatment consists of giving injections of antivenin, often intravenous fluids, and frequently cortisone. Antibiotics and tetanus antitoxin are also frequently given to prevent further problems once the bite has been treated.

Strychnine Poisoning

In the past strychnine poison was used to control rats and mice. While it has been replaced with such products as warfarin, it is still found in many rural areas for use on gophers and prairie dogs. And there are some malicious people in the world that use strychnine to poison neighborhood dogs.

A poisoned dog will usually begin to show symptoms an hour or so after having eaten the tainted food. It will begin to tremble, pant, and act nervous. As the symptoms progress, it will respond to any loud noise, such as a car horn beeping, by having convulsions. As the condition worsens, even clapping of the hands can produce a severe convulsion.

If you take the dog to a veterinarian immediately, it can be given emergency treatment. This consists of emptying the stomach to prevent further absorption of the strychnine, intravenous injection of sodium pentobarbital to control seizures, and often an intravenous dextrose solution to provide fluids and improve kidney action.

When left untreated, the dog will suffer one convulsion after another until it dies from exhaustion. It is a cruel death that not even a rat should suffer.

Ticks

Ticks are quite a problem for dogs in many areas. There are several types of ticks that infest dogs; all are basically the same. They are blood-sucking parasites that attach themselves to an animal until full of blood. In small numbers they generally cause no trouble except irritation. In large numbers, however, they can produce anemia and extreme agitation. Scratching and digging at the tick bites break off the tick heads, irritate the skin, and aggravate the situation. Secondary infections often follow. Unfortunately, in increasingly large areas of the country, ticks can carry Lyme disease, which is contagious to humans. This disease is often carried by the brown deer tick, which can be very tiny. (For more information, see "Lyme Disease" on page 324.)

It is wise to examine your dog daily for ticks. Remove these pests with a

gloved hand and a ball of cotton or tissue. Pull slowly but firmly until they release, and then burn them to be sure they are completely destroyed. If you just yank a tick off, the mouth may remain attached, causing infection and soreness.

Where ticks are a problem, there are topical flea and tick killers that are very effective. These, such as Frontline, available through your veterinarian, are applied in a strip down the back, on the skin. They are then absorbed into the body and bloodstream making the dog's blood toxic to biting insects. Your dog may get ticks on him but when they bite, they will then die and fall off when quite small. It is still a good idea to check your pet daily for ticks.

Tonsillitis

Tonsillitis occurs often in dogs. It is usually caused by a bacterial infection. I have seen many dogs having a simultaneous anal gland infection caused by the same bacteria infecting the tonsils. I believe these dogs had their anal gland infection first, licked the anal area in an attempt to alleviate the condition, and picked up the infection in the tonsils. In chronic cases, once the anal glands were removed, the dogs never had tonsillitis again. Coincidence? I hardly think so.

A dog affected with tonsillitis will usually cough, gag, and vomit. It may paw at its throat. Often there will be a fever, lack of appetite, dullness, and lack of energy. Many times you can feel one or both tonsils, just below the jaw, when they are swollen.

If you suspect tonsillitis, consult with your veterinarian. When taken care of promptly, most cases of tonsillitis clear up in three to four days. Using a broad-spectrum antibiotic, plus cortisone to take the swelling out of the tonsil and throat, works in most cases. Good nursing also plays an important part. You can often coax your dog to eat by giving it ice cream, cold milk, and cold broth. These foods are soothing on the sore throat and will encourage the dog to take more nourishment.

With cases of recurrent tonsillitis where anal infection has *not* been present, it is often advisable to remove the tonsils surgically.

Worms

Every year your dog should receive a regular fecal examination as well as a heartworm check, done via blood sample, by a veterinarian. If there are worms present, your veterinarian can then suggest the right wormer for the job. True,

there are “shotgun” prescriptions available at many stores that claim to kill all worms. Unfortunately, they either do not kill the worms or, worse, can kill the dog if misused. Each worm requires different drugs to kill it. Some drugs kill several species, some only one. Few wormers do a good job on more than three types. Therefore, examination and treatment by your veterinarian are essential to keeping your dog worm-free.

Heartworm

Heartworms infest the heart. Once mainly a problem in southern areas, heartworm has become widespread. It is spread by mosquitoes, which suck the blood of an infected dog and transmit the microfilariae (prelarvae) to another healthy dog. Here they develop into mature heartworms. These in turn reproduce, and there are more microfilariae to infect another dog. Heartworms are large worms and can completely block the heart. When a badly infested dog is exercised, it may show shortness of breath, coughing, and lack of stamina, and it may be easily tired. Sudden death can result.

Every dog should have a yearly check for heartworm. If found free of them, it can be put on an oral heartworm preventive. This is given throughout the mosquito season.

Treatment for heartworm can be tricky but is usually successful. The drug that kills the adult heartworms must be used with extreme care: If too many worms are killed quickly, they leave the heart, causing blockages in the bloodstream, like a spring logjam. This can result in death. Often, repeated treatments are necessary. Following treatment the dog must receive very restricted exercise until your veterinarian feels the adult worms have all been killed and passed out of the heart and lungs.

Hookworm

Hookworms are one of the most severe intestinal parasites. They cause unthriftiness, diarrhea, blood in the stool, and death in young puppies due to anemia. Puppies can be infected before being born and become reinfested very early. Death may result at an early age. Hookworms enter the body by penetrating the skin. In this way they can also infect people, especially those who prefer to walk barefoot.

All puppies should receive a fecal examination for worms at weaning; but if there has been trouble with hookworm in the past or if the pups are not doing as

well as they should, they should receive an examination at two to three weeks of age. If you wait until eight weeks, there may not be any puppies left.

Strict sanitation measures are necessary when hookworm has shown up. As the larvae are in the soil, you should move the dogs to a new, clean area after worming them. Have regular stool checks done after that to ensure that the parasite is really gone, not just in hiding. There are chemicals that kill hookworm larvae on concrete runs and in the soil, to some extent; ask your veterinarian for his advice.

Roundworm

Roundworm is the most common parasite affecting dogs. Most puppies are born with roundworms, even when the dam was faithfully wormed before she was bred. These worms are quite large, white, and tapered at both ends, resembling a bean sprout. They are often coughed up or seen in the stool, especially in puppies. A small puppy can harbor huge amounts of these parasites, which harm the digestive tract and cause malnutrition. (After all, you can't feed a puppy what it needs if the worms beat it to the food!)

The worms migrate through the bloodstream and thus can be difficult to treat. One worming is not enough. That kills only the worms present in the stomach and intestine, but it does not kill the larvae in other places or destroy the eggs that will hatch and soon reinfest the dog. Worming a second time, two weeks after the first treatment, will often provide best results. Ask your veterinarian for his suggestions.

Tapeworm

Tapeworms are carried to the dog by fleas. A flea larva ingests tapeworm eggs, becomes an adult, and is ingested by the dog; then the eggs hatch, starting the cycle again. The adult tapeworm is large and flat and can grow to a length of several feet. It is made up of segments, each containing eggs. These sections are passed in the stool; they are whitish pieces that look like inchworms when first passed and quickly dry to look like grains of rice. They are often found in the hair around the dog's rectum and tail or in its bedding.

Your veterinarian can prescribe a drug that will effectively rid your dog of tapeworm. Along with fleas, rabbits also serve as an intermediate host of the tapeworm, so never feed rabbit entrails to your dog. Reducing the flea population on the dog and restricting its diet to domestic foods, such as dry dog

food, as opposed to “what it catches,” will help keep it tapeworm-free.

Whipworm

Whipworm is less common than the other worms. It can be hard to get rid of, though, because it lives in the colon and cecum (a blind pouch resembling the appendix in a human). Often the most common sign of whipworm is the liquid, foul-smelling stool. Unthriftiness and anemia follow.

Consult your veterinarian for diagnosis and a treatment program based on your dog's needs. Treatment consists of giving oral drugs or occasionally, where warranted, injectables. Strict sanitation will do much to cut down on whipworm infestations, as well.

CATS

GENERAL CARE AND MANAGEMENT

Although the average cat is very independent, it does require care to remain happy and healthy. Daily feeding isn't enough; all too many cats die at an early age owing to their owners paying no attention to their other needs or cat comforts.

Housing

The house cat should have a clean litter box available or have use of a pet door (in safe rural areas) to go outdoors when necessary. A scratching post, for the natural process of cleaning the claws and for exercise, is appreciated both by the cat and its owner; without it, the cat will often use the furniture or carpet. Likewise, a few toys, such as a sock stuffed with another one, a catnip mouse, a ball, or some of the more sophisticated and expensive toys, will make your cat happier and healthier as it will get more safe exercise.

Although a cat may nap on the sofa or chair, most appreciate their own beds. A simple cardboard box with a couple of old towels in it will work well, or the owner may opt for a more elaborate "kitty condo," carpeted and made fun with cutout windows throughout.

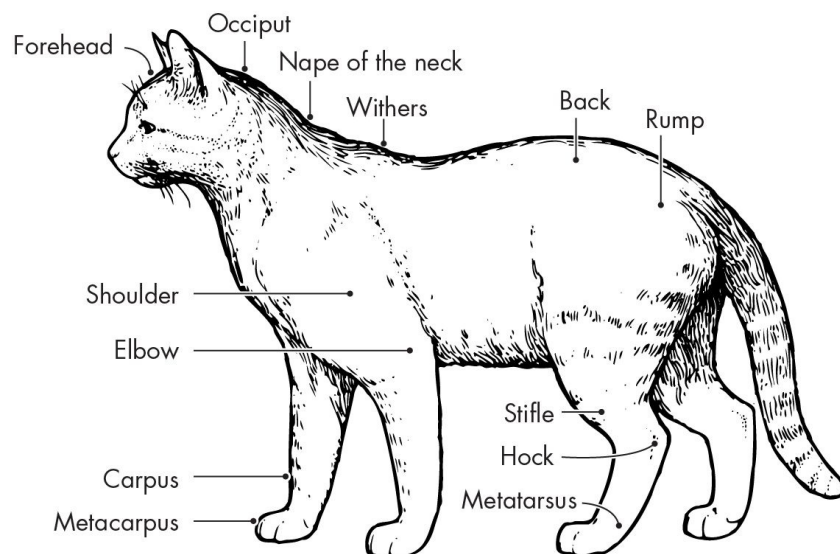
Feeding

Generally, cats do well with a good-quality dry food available at all times, with perhaps a small can of food given daily as a treat. I stress "good-quality" here. Several brands of cat food are cheaper than others, but they are also lower in quality, having lesser ingredients than the better foods. You can usually trust your cat to tell you. If it likes the food, it is often of higher quality than one the cat turns its nose up at. Shop around, buying small bags at first, until both you and your cat are satisfied. Reading product labels is a help, but it is getting increasingly hard for the layperson to understand terms such as "poultry digest"

and “meat and bone byproducts.” Again, trust your cat. If it appears sleek, active, and happy, you’ve made the right choice. If it is rough coated and sluggish, has bouts of diarrhea, or vomits its food, make a switch.

Do not feed cats any cooked fish or poultry containing the bones. Small, sharp bones can cause not only stomach pain but also impaction of the intestine and death. Also, avoid feeding highly seasoned foods to cats. Remember, nature did not intend cats to eat hot peppers, bologna, or sausage. Gastric upsets often result, which can lead to more serious troubles. Keep in mind, too, that cats do not like sour or moldy food. Never leave moist food, such as canned cat food, in a bowl for longer than six hours. If it is too much for the cat to eat in that period, feed less food. Sour or moldy food can be filled with bacteria that will make the cat sick or even kill it.

PARTS OF A CAT



First Aid Kit for Cats

Here’s a list of suggested supplies for your feline first aid kit. Every home with a cat should have at least the items marked with an asterisk. If you have several cats, all of the items in this list may

come in handy at some time. Ask your veterinarian for his recommendations for your particular locale and situation.

- Antiseptic wound liquid*
- Cat carrier*
- Cotton*
- 5-cc syringe
- Forceps*
- French feeding tube
- Pet nail clippers*
- Pet nursing bottle
- Rectal veterinary thermometer*
- Veterinarian's phone number*

If you keep cats to control mice, don't depend on their hunting to supply the food they need. The old idea that a hungry cat is a better mouser is untrue. The well-fed, but not fat, cat gives the best in its hunting. A female cat, hunting instinctively for her kittens, will hunt all day. She should have adequate nourishment to do this. Her kittens should also receive additional food as often one aggressive kitten will grab all the prey the mother brings, leaving its brothers and sisters hungry.

All cats should always have access to fresh, clean water. If they do not receive adequate water, they may become dehydrated and susceptible to illness. Inadequate water consumption, especially with a diet high in minerals and ash, can lead to a problem known as urinary calculi. Dry cat food can also cause trouble in this way. If the cat does not drink adequate water to compensate for the lack of moisture in the dry food, it is more likely that urinary calculi will form. (For more on preventing and controlling this problem, see "Urinary Calculi" on page 362).

Do not let your cat drink out of the bathtub or toilet bowl, because many cats are poisoned by drinking toilet bowl fresheners, Lysol, and cleaners.

Restraint

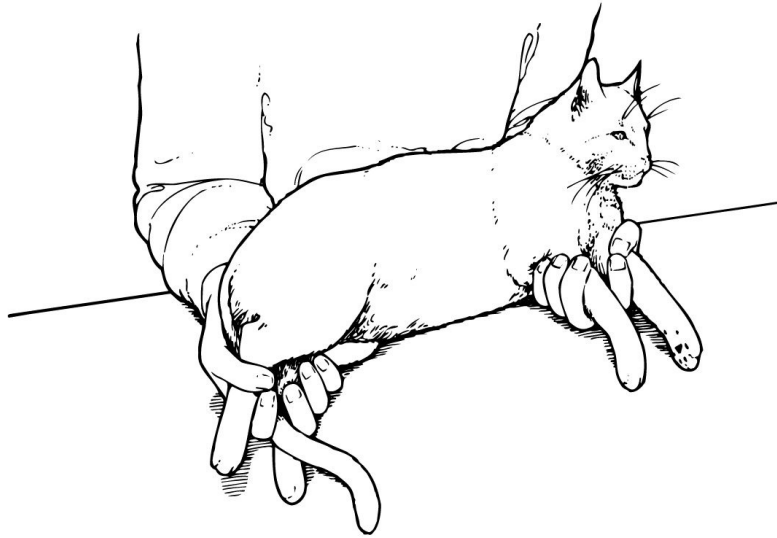
Occasionally it will be necessary to restrain your cat. This is much more

difficult than restraining a dog as cats are generally more “wiggly” and aggressive in fighting restraint. Using tried-and-true methods of restraint will make it much easier for you or your veterinarian to examine, treat, or groom your cat.

Sometimes there is a certain knack to just holding onto a cat. When it has decided that it would rather be elsewhere, a cat can turn into a slippery, twisting bundle with claws. Being a naturally nervous animal, a cat can be upset by many things, even if it normally has a calm disposition. A loud noise, a moving car, barking dogs, and strange surroundings can all bring on the desire to “escape” and hide. The best way to hold a cat that wants to wiggle away is to grasp the forefeet with one hand and the hind legs and tail with the other hand. The cat uses the tail for balance when jumping and the hind legs for ripping and clawing when frightened or angry. Restraining both front and hind legs, plus the tail, will enable the average person to hold on to all but the most frightened or vicious cat.

This hold, however, gives no protection to you against bites. Most cats will not bite unless hurt first. When given an injection, though, some will try to claw or bite. The easiest way to hold the cat for a procedure such as this is to grasp the skin firmly behind the neck and just in front of the tail, placing the cat on a slippery table or other surface. Use care not to pinch the skin, and hold the cat’s body close to your own for security and added restraint.

Wrapping the cat in a towel sometimes aids in minor treatment but will not work with a very frightened cat. When you need more restraint, slip the cat into a canvas bag with a drawstring at the top, allowing the head to come out. Then gently close the bag snugly (but not too tightly) so there isn’t enough room for the front legs to work free. This restraint works well for medicating ears, cleaning teeth, giving oral medicines, and examining the head of an excitable cat.



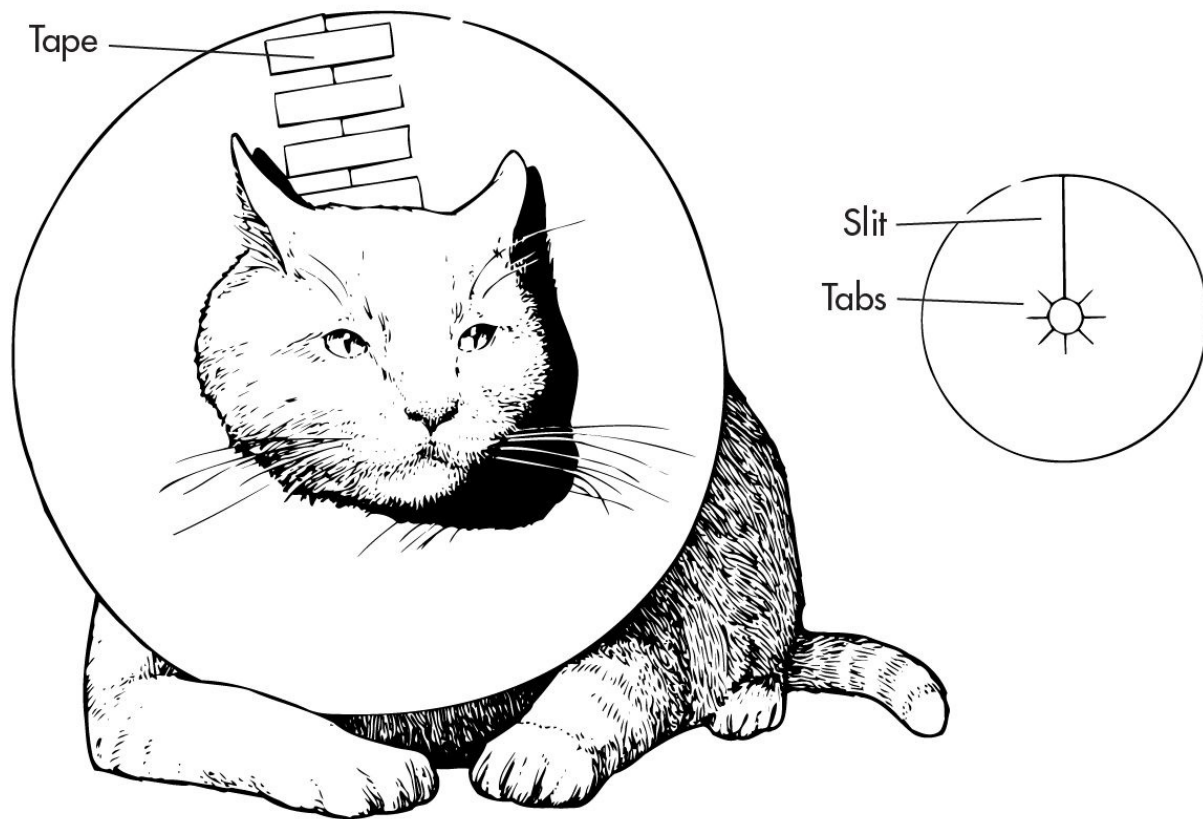
Restraining a cat. *If your cat is generally calm, you can restrain it by grasping the forefeet with one hand and the hind legs and tail with the other. This method will not prevent bites, however.*

Once in a while, it may be necessary to prevent your cat from scratching at its ears, rubbing its eyes, or licking medication off its body. Although tranquilizers can sometimes prevent this, many a cat will continue to scratch or lick, even though really drugged. It might not hit the right place, but it will continue to try, sometimes causing damage. The easiest way to keep the cat from bothering medication or to prevent self-mutilation due to an itching problem (such as ear mites) is to make an Elizabethan collar. A variety of materials will work, but the most common are plastic, such as that cut from plastic pails, and heavy cardboard. Make the collar in the shape of a circle, perhaps 12 inches in diameter, with a hole in the center that fits snugly around the cat's neck. Then make a cut in the collar, from the neck hole to the outside rim. Fit the collar on the cat, overlapping the cut edges and taping them together to form a cone-shaped arrangement. When a collar is the right size and well fitted, only the most determined cat will be able to get it off or get around it with feet or mouth.

Breeding

The queen, or female cat, will come into heat every two to three weeks, most often in the winter, spring, and summer. She is in heat for a period of three to six days. Signs of heat are an arched back, with hindquarters high in the air, a throaty, yowling call, rolling about on the floor, holding the tail upward, and

treading the hind legs.



Elizabethan collar. To prevent a cat from licking a wound or medication, create an Elizabethan collar out of plastic or heavy cardboard. Make a slit in one side, and then cut out a smaller circle in the center. Cutting and bending back tabs around the center circle will make it harder for the cat to remove the collar.

Before allowing your cat to be bred, consider that multitudes of unwanted, mistreated stray cats are already on the streets and being destroyed by the hundreds every day at animal shelters for lack of homes. Do you really want to contribute to that problem by allowing your queen to have a litter of perhaps eight kittens? It is extremely hard to find good homes for kittens, even well-bred purebred kittens. It is not good enough to say, "I'll give them to a farmer." Most farmers have too many cats already! Really think it out, and be *sure* you have available homes for all kittens, even ones you can't find other owners for. (This means you'll probably be stuck with one or more female kittens, who will come into heat in a few months and possibly become bred themselves.) It is much better to spay your female pet than to allow her to raise a litter. Spaying will not

change her disposition, make her fat, or make her uninterested in hunting. If you want a kitten, find an unwanted one to raise.

When you definitely desire to breed your cat, plan the breeding with care. The queen should be in the very best of health. Have her checked for worms and other parasites before she is even due to come into heat. She should carry immunity to the diseases she can be vaccinated against. This will not only protect her but also give immunity to her kittens from birth to about eight weeks of age in most cases.

Make arrangements with the stud owner before your queen is due to come into heat. If she is to be bred to one certain male, keep her isolated in a secure room to prevent an accidental mating with another male cat.

Following a successful breeding, the queen will often seem to suddenly go out of heat. This is because copulation stimulates ovulation.

Pregnancy can often be detected at about three weeks by an experienced person. Your veterinarian can perform this examination for you. When the abdomen is gently palpated, marble-sized lumps in the uterus can be felt. Do not squeeze the abdomen roughly or very hard as damage to the queen or the kittens can result.

The pregnant queen should receive a balanced diet and be encouraged to eat high-protein foods, such as liver. Also, give a good vitamin-mineral supplement to aid the developing kittens and the queen. The mother's appetite will increase with the size of her abdomen, and she should receive plenty of quality food. It is natural for her to slow down and not play or run about as she may have done prior to feeling her pregnancy.

Birthing

As the queen's time draws nearer, she will seek out a quiet, secluded "nest," such as a dresser drawer, a hole in the hay, or a place in the closet. Fifty-eight to 68 days after breeding, she will be ready to give birth to her kittens. Her temperature will drop sharply 12 to 24 hours preceding birth. The kittens are usually born with no trouble, owing to their sausage shape and small size, but sometimes a physical defect in the queen or other problem will hinder birth. If she labors for longer than half an hour with a kitten, producing nothing, call your veterinarian. He may advise a little more time, an injection to help uterine contractions, or possibly a cesarean if it is impossible for her to make a natural delivery.

Because of the very small size of the vagina and the pelvic opening in the queen, it is best to take her to the veterinarian at once when there are birth difficulties. A queen tires quickly in a difficult birth and can be easily injured by inexperienced attempts to help her. Many a uterus has been ripped by too forceful an extraction of a fetus, and many a queen has been lost due to shock and exhaustion.

If the difficulty is just a larger kitten becoming stuck at the vaginal opening, this is generally no problem. Grasp the head or legs with a dry towel, and pull gently but firmly as the queen strains. The kitten usually is then quickly expelled. Do not just yank the kitten out. This will cause damage and bleeding.

In most cases, however, the kittens are born easily. It is a good idea to keep an eye on the queen while she is having them, for you can often help save a kitten. Once in a while, a queen may be busy cleaning one kitten, ignoring the fact that she just had another one. If the membrane is not removed from the kitten's head after the umbilical cord breaks, the kitten will often smother or drown in fluid. If the queen does not remove the sac, you should do it for her, drying the head (especially the nose and mouth) well.

Each kitten is followed by a placenta (afterbirth). Count these to make sure there is one for each kitten. Once in a great while, a queen will retain one or more placentas, causing an inflammation and infection of the uterus. If you are sure there is a retained placenta, call your veterinarian. Often an injection of hormones is all that's needed to encourage the uterus to release the placenta. Don't, by the way, be alarmed at the color of these placentas: They are often a horrible greenish black. Also, don't be alarmed when the queen eats them; this is normal. Eating the placentas stimulates her milk letdown.

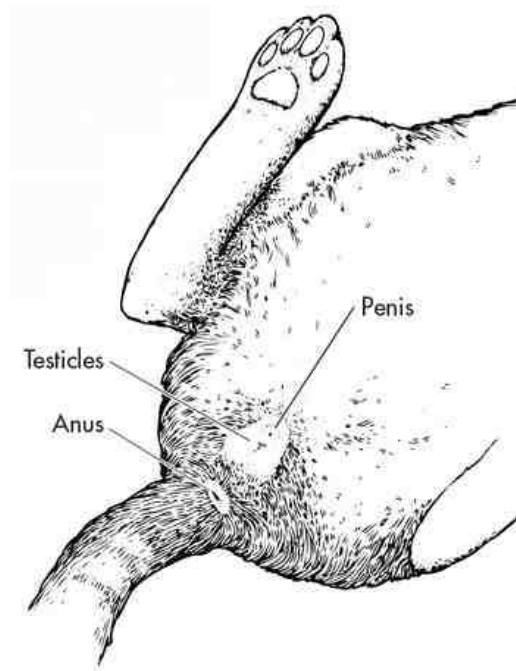
Caring for Kittens

The kittens will be born with their eyes shut tightly. These will open in about two weeks, and the kittens will begin to move about.

It is easiest to sex your litter of new kittens just after birth, if desired. Right after birth, the testicles of the male kittens are readily apparent. In a day or so, it can be very hard to sex the kittens unless you are experienced. Hold the newborn kitten on its back. In the males you will see the testicles as tiny bulges just below the anus, on either side of the penis.

The kittens will nurse for six to eight weeks of age, but as soon as they will eat semisolid food—at about four weeks—you should give them several small

feedings daily to relieve the queen of the strain of producing milk.



Sexing a kitten. *It's easiest to determine a kitten's sex right after birth. On males the testicles will be readily apparent as tiny bulges just below the anus, on either side of the penis.*

Kittens can be weaned at eight weeks of age if you are prepared to give them adequate care. (Otherwise, leave them with the mother until she weans them herself or her milk dries up naturally.) Small kittens should be fed four or five times daily. When they are about three months old, cut down to three meals a day, and at six months, feed them twice daily in addition to providing free-choice dry food of a good quality. Kittens need a 30 percent protein diet, so feeding a dry food with 14 percent protein will not maintain a kitten or allow it to grow. Inadequate protein intake will usually result in death. Feeding such foods as liver, kidney, raw ground beef, and small amounts of milk will boost the protein intake, as will feeding a good canned cat food made up of such things as beef and liver. Too much milk or cereals (as found in cheap cat foods) will cause indigestion and diarrhea.

Orphan Kittens

Once in a while, either due to lack of milk on the queen's part or due to her

untimely death, you may need to raise the litter by hand. This is quite a job, so be prepared! Fortunately, it is also very rewarding.

The kittens should get about 2 cc of milk (preferably goat milk) every two hours, day and night, for the first week. If the kittens seem very hungry after feeding, increase the amount slowly until they are a little hungry, but not mewling continually, after feeding. It's easy to give the milk, either by tube-feeding (see "Orphan and Rejected Puppies" on page 297) or by using a pet or doll bottle with a soft rubber nipple. An eyedropper is not as good, because it gives the milk too fast, causing inhalation pneumonia as the milk droplets enter the lungs.

After the first week, increase the milk to about 4 cc, depending on the size and health of the kittens. Kittens being fed the right amount of milk will feel full and round, with tight skin. They will squirm and mew angrily when hungry. Too much milk will cause diarrhea, which is often fatal.

As the kittens' eyes open, gradually increase the amount of milk while decreasing the number of feedings per day. At three to four weeks, add semisolid food to their diet in the form of mashed liver.

One caution here: If the queen cannot be around to take care of cleaning the kittens, you are elected—not just for pretty kittens but for their very life. Kittens need to be stimulated to urinate and defecate for a week or more after birth. The mother does this by licking the genital areas. You will have to use a slightly damp wash cloth or tissue after each feeding. If this is not done, there will be serious gastric troubles, and often the kittens will die.

Neutering

Any pet cats, especially those kept in the house, should be neutered—unless, of course, you plan to use them for breeding.

Castration

An uncastrated male will begin to spray urine (which doesn't smell like cologne!) about the house, marking his "territory," at about 9 to 12 months of age. It is the rare male that does not do this. The odor is lingering and hard to remove from fabric, such as rugs and drapes. The unneutered male will also begin roaming, seeking romance. Unfortunately, other toms will have the same idea about the same females, and fighting results. The pet male often returns from these brawls scratched, bitten, and torn. Infections are common.

On a male about 8 months old, this quite inexpensive and safe operation will end these problems before they begin. Castrating an older male who is set in his ways may slow him down quite a bit but may not completely alter his spraying and fighting habits.

Spaying

With an unneutered female, there are other problems. Once every two to three weeks, she will be in heat, rolling about and yowling constantly. Trying to find a place to put her out of hearing, but still away from the visiting males, is a very real problem. If you leave the door open, she will be out like a flash, presenting you with a litter of two to eight kittens in about two months.

The spaying procedure is briefly outlined in “Spaying” on page 299. The same procedure used for dogs is generally used on cats. The operation is usually safer with a cat, however, due to the smaller size of the cat’s uterus and blood vessels. If possible, have your cat spayed between 5 and 10 months of age. It is not necessary or desirable for her to have been in heat or to have had a litter before spaying her. She will not suffer from being deprived of motherhood but will be spared the chance of having uterine infections, undergoing birth difficulties, and experiencing the stress of bearing and raising a litter.

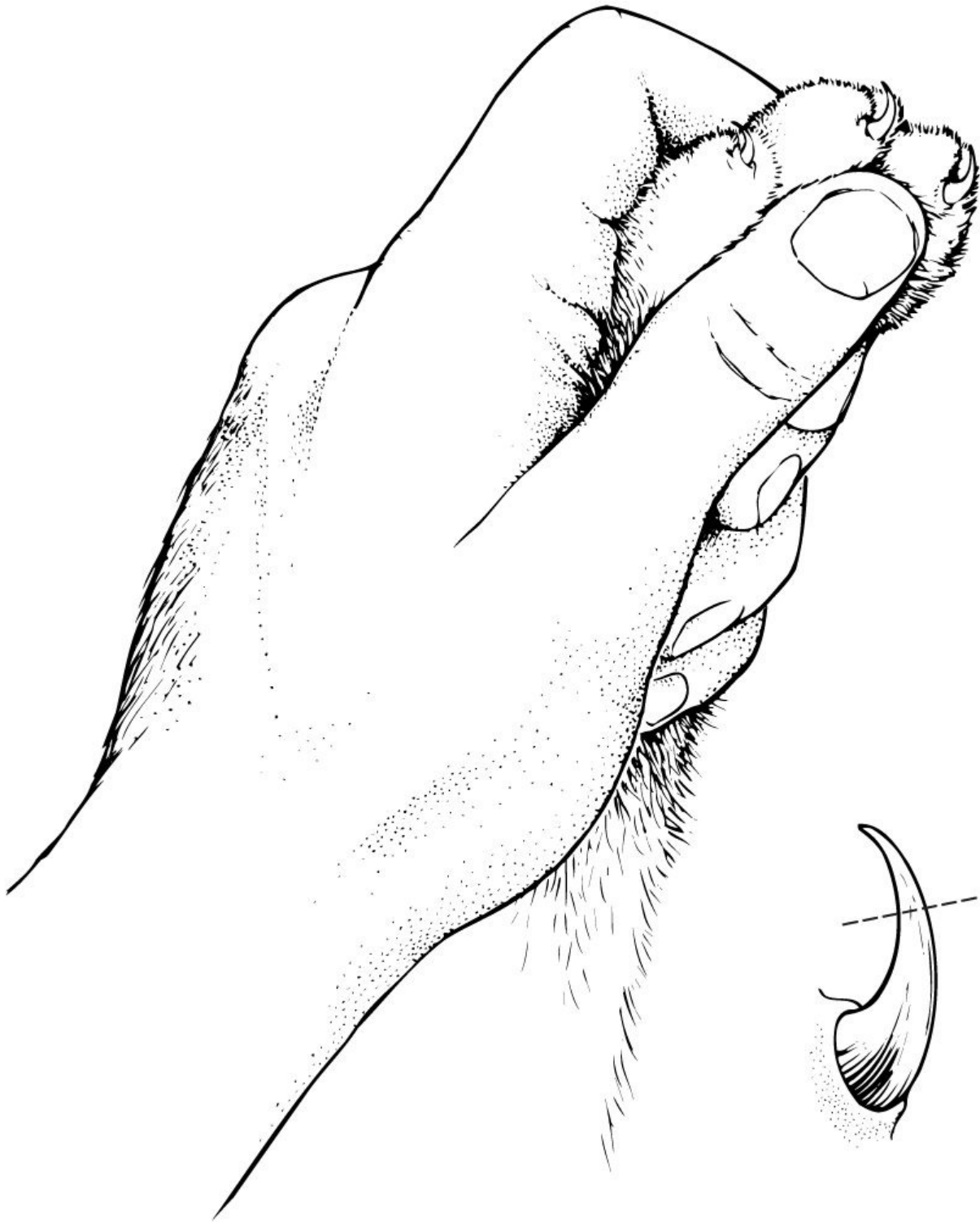
Grooming

Regular grooming is very enjoyable to your cat. It will also help prevent such health problems as hair balls, caused by the cat ingesting hair while grooming itself. For short-haired cats, all you need is a soft brush. A pin brush (with straight metal pins) or slicker (with bent metal pins) is usually necessary for longhairs. Brush gently, with the lay of the hair. A finishing rub with an old towel feels good to the cat and will make the coat shine.

Trimming Claws and Declawing

Due to the extreme sharpness of a cat’s claws and the tendency for some cats to use those claws on people or furniture, owners often want them dulled or removed. There are two choices here: declawing, which is the surgical removal of the claw and first joint of each toe, or trimming the nails on a regular basis. I would recommend trying to trim the claws before deciding to declaw the cat. Declawing is permanent and does remove the cat’s prime defense. A declawed cat can hunt and climb trees but is hampered compared with its clawed

counterpart.



Trimming a cat's claws. Press upward between the toes to unsheathe the claws. Using a pet nail trimmer, clip off the hook at the end of the claw. Avoid cutting into the pink blood vessel that runs partway down into the claw.

A cat is usually easy to work on when trimming the claws as it causes no discomfort, and you can do a few at a time. The cat's claws are retractable, so you have to press upward between the toes to unsheathe the claws. Most cats have light-colored claws, making it easy to see the pink blood vessel that runs partway down inside the claw. Using a pet nail trimmer, clip off the hook on the end of the claw, staying away from the blood vessel. If you do nick the vessel, touching a styptic pencil to the end will stop the bleeding. But cutting so deeply hurts the cat, and it will not be so trusting in the future. Clip one claw at a time until all four feet are finished. The hind claws are not as long or as sharp as the front claws, so do the claws on the front paws first.

If you feel it is necessary, your veterinarian can declaw your cat. But with regular trimming, most cats do not require this surgery to cease clawing their owners and the furniture.

Caring for Teeth

Cats are often prone to dental problems, especially in their later years. A soft diet, such as canned cat food, does not massage the gums or clean tartar deposits from them. These tartar deposits build up and irritate the gums. Swelling, bacterial infections, and abscesses are often a result.

A cat that drools, either when eating or just resting, should be examined for tooth and gum problems. Foul breath is also a sign of dental trouble. Often a cat with bad teeth will eat hard food (such as dry cat food) with its head cocked sideways as it tries to avoid chewing on the sore tooth. Likewise, a cat with an abscessed tooth will often paw at its mouth and shake its head.

To catch problems early, check your cat's mouth periodically, and take it to the veterinarian if you notice any redness of the gums, foul breath, or teeth with brown tartar deposits. If the teeth are brownish or yellow, they should be cleaned. If there are only a few spots of tartar, you can remove them at home by scraping the tooth with the edge of a dime. Few cats will allow a thorough cleaning at home, however. It's easiest to have your veterinarian do this, using a general anesthetic.

Giving your cat dry food from time to time will aid in preventing tartar

buildups and gum troubles because it keeps the teeth scraped clean. Likewise, there are special kitty treats formulated to help keep tartar to a minimum.

Transporting

It is sometimes necessary to take your cat some distance by car. By nature, few cats enjoy riding in a car, and the motion and noise make them nervous and possibly even sick. (Of course, if a kitten is taken for rides from an early age, it is *much* more apt to enjoy riding in a car when it is older!) The first reaction of the cat is to try to escape; if this fails, the cat will usually hide under the seat, becoming almost impossible to extract when you want to get it out. And a frightened cat is apt to dash out the door the instant it is opened, never to be seen again.

For this reason, the safest means of car travel for a cat is inside a sturdy cat carrier. It is a great convenience and a great protection to a cat, making accidental escapes nearly impossible. A cat likes to feel hidden and secure, so it will frequently ride happily when snuggled in a cat carrier. Carriers are getting very inexpensive to buy, and I feel every cat owner should own at least one.

For longer trips than to your veterinarian, a larger carrier is necessary. It should allow room for a small litter box besides the cat and its curl-up bed.

Sample Vaccination Schedule for Cats

Keep in mind that this is a suggested vaccination schedule. Your veterinarian may use different products or suggest a different schedule based on your location and situation.

	DISEASE	WHEN TO VACCINATE
KITTENS	Feline distemper, feline leukemia, pneumonitis, rhinotracheitis, and feline calicivirus (combination)	9 weeks; booster at 12 weeks

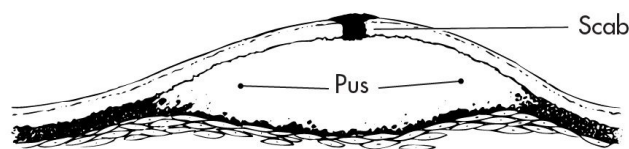
	Rabies	6 months
CATS	Feline distemper, feline leukemia, pneumonitis, rhinotracheitis, and feline calicivirus (combination)	Yearly
	Rabies	Yearly

DISEASES AND OTHER PROBLEMS

Although cats are generally very healthy and disease-free, it is wise to learn all you can about common diseases and health problems. This way, you'll recognize symptoms of possible trouble before the problem gets serious. It's also important to know the normal temperature of your cat so you can check for fever or other abnormalities. The normal body temperature of a cat is 101° to 102°F (38.3° to 38.9°C) .

Abscesses

Abscesses are common on cats, particularly males. Fight wounds, such as bites and scratches that puncture the skin but do not tear, often become infected and then close up, sealing bacteria under the skin. Such punctures will later abscess. With the normal loose skin of the cat, the abscess can spread under the skin and fur until all of a sudden a large bulge becomes noticeable. The abscess may break and drain, but often it will ooze through perforations in the skin and not drain adequately.



Cross section of an abscess. Before lancing a suspected abscess, your veterinarian will withdraw some of the fluid. An abscess contains yellowish or whitish pus.

This abscess should be lanced near the bottom with a sterile instrument. It is safest to have your veterinarian do this, with the cat under an anesthetic, so the area can be thoroughly flushed and cleaned. The abscess should be kept clean and open until it is healed from the inside. Antibiotic therapy is usually advised to prevent a bloodstream infection.

Never be tempted to lance a bump yourself unless you are positive it is an abscess. Hematomas (blood-filled bruises) look like abscesses, but if they are lanced, the cat may bleed to death. Therefore, it is not wise to attempt home treatment of an unopened abscess. Your veterinarian will first withdraw some of the lump's fluid with a needle and syringe to make sure it is yellowish or white (indicating an abscess) instead of dark or bloody (indicating a hematoma).

Breaks

Legs are the most often broken bones in the cat. Maybe this is because they are long and thin and they are positioned in such an unprotected way that they tend to get caught and wrenched. The tail is another area that frequently suffers breaks, usually due to having been slammed in a door.

A cat with a broken leg will commonly limp very badly, holding the leg off the ground. The leg will usually dangle unnaturally. If you suspect a broken leg, quietly get hold of the cat, and carefully place it in a secure cat carrier padded with an old towel. Many injured cats desire privacy and a hiding place when the initial shock wears off. Call your veterinarian at once. A broken leg is most successfully set the day it is broken—not a week later. Cats that are seriously injured and left untreated can develop complications and possibly die.

Do not attempt to put a splint, even just temporarily, on the cat. Many serious complications have arisen following improper placing of a home splint. It is better just to allow the cat to remain quietly in a carrier than to place a splint that may cause more damage and upset the cat.

Most broken legs are easy to set on a cat, with an intramedullary pin, a light aluminum or plastic splint, or a light plaster cast. Cats are generally good patients and heal quickly with good nursing.

Some broken tails can be set easily at home by bringing the tail into a normal position and taping strips of stiff cardboard along the tail to hold it in place. Avoid taping tightly, or you could cut off circulation to the area; the tail will become necrotic and fall off. If the break is near the tip of the tail or if you are nervous about treating the break, take the cat to your veterinarian. He can often

splint the tail successfully. Otherwise, a broken tail will often heal by itself, but the cat will permanently carry a kink or two in its tail.

Car Accidents

More cats are injured by cars than any other way. Because cats are hunters, it is natural for them to seek out the grassy roadsides to hunt on, often at night. Unfortunately, many times they are frightened by traffic and then panic, run onto the road, and are hit. If you notice that an outdoor cat suddenly acts as if it is in pain or if it is lame or listless, suspect that it has been hit by a motor vehicle. Inspect the body closely for abrasions, bald spots, bruises, or swellings and sore spots. Look at the gums, as pale gums indicate either shock or internal bleeding. Also, check the eyes to see if one pupil is more dilated than the other, indicating a concussion. Where both are dilated, there may be only shock, or possibly there is a head injury. If you see any signs of possible injury, take the cat to your veterinarian for a thorough examination and perhaps observation for a day or so. Sometimes seemingly minor injuries can worsen over time, with bad results.

Should you see your cat struck by a vehicle, quietly capture the cat, gently place it in a cat carrier, and immediately take it to your veterinarian. Waiting even a few hours could cause death.

Cats also seem to have a problem with fans under the hoods of cars. They like to get up under the hood, especially in cold weather, and when the motor is started, they get caught in the fan belt. If this happens, do not panic. Place the cat in a quiet, dark carrier with some light bedding, and drive to your veterinarian at once. A seemingly horribly torn-up cat may live and make a complete recovery with veterinary help. Don't listen to well-meant advice to "put the cat to sleep" unless it comes from your veterinarian. I've seen some really cut-up cats recover in a short time—no scars even! Cats have a good healing ability when given a chance and good nursing.

Car Sickness

Cats, on the whole, are not good travelers in the family car. They often are not introduced to traveling in a vehicle until they are adults, and the motion frightens them. They may get extremely nervous at the sight and feeling of the movement of the car. They can get so frightened they can go berserk, clawing and yowling. More frequently, they become carsick, hiding under the seat, yowling, drooling, and sometimes becoming nauseated.

Often car sickness can be avoided by placing the cat in a dark cat carrier. Feeling hidden and more secure, the cat will not be as nervous or get carsick. You can also give the cat a light tranquilizer a while before the trip, but sometimes the drug has the opposite effect than you had hoped for. If you plan to tranquilize your cat, it is best to try out the tranquilizer on the animal while at home, before the trip.

If your cat is going to have to ride periodically, it is a good idea to get it slowly used to riding. Short trips, such as going to the corner store, school, or park, are easier on the cat's nervous system, and soon it will nearly relax in the car. Be sure to use a cat carrier as having a cat loose in a vehicle is dangerous: Cats, especially frightened cats, can dash out an opened car door in a flash, often never to be seen again. Slowly increase the time your cat spends riding, until it pays no attention to a trip. Beginning this at a young age makes even a long trip a relaxing, everyday happening.

Conjunctivitis

Conjunctivitis is the inflammation and often reddening of the conjunctiva (the light pink lining around the eyeball). Simple conjunctivitis is often caused by dust, weeds, pollen, or other foreign bodies irritating the area, causing inflammation. Tearing and squinting are commonly found in this problem. Bacterial infections often follow the initial irritation, causing even further irritation and swelling. The cat may rub its eye with a paw or rub its head on the floor. This serves only to irritate the area further.

Most cases of simple conjunctivitis or conjunctivitis complicated with a bacterial infection will clear up when you use an antibiotic ointment for a day or two. There are some inexpensive but excellent ointments on the market now. Besides containing an antibiotic, they also include a local anesthetic to deaden the pain and itching.

You should be aware, however, that conjunctivitis is often a symptom of a disease, such as feline calicivirus or rhinotracheitis. So, if it does not clear up with minimum home treatment or if the cat acts sick or runs a fever, take your cat to your veterinarian at once.

Constipation

Constipation is often associated with improper diet. Eating small bones or eating dry food without drinking enough water will often cause constipation.

Never feed any small bones, such as fish, chop, or chicken bones, to your cat. Besides causing constipation, they can also puncture the intestines and kill the animal. When feeding a dry cat food, alternate it at least twice a week with good-quality canned food. If the cat is frequently bothered by hard stools, it may be wise to put about 1 teaspoon of vegetable oil in its food daily or to feed more canned food. Adding milk to a dry diet will help prevent constipation, too.

Adequate exercise is also very important in keeping the cat regular. A cat that just sits in one place all day is prone to constipation. Two other possible causes of constipation are hair balls and a fever, which often partially dries out the intestine, making hard stools appear.

If your cat frequently suffers from constipation, have your veterinarian give the animal a checkup. If there is no health problem, he can prescribe a simple stool softener to be added daily or periodically to the cat's food. These are quite effective and provide great relief to the cat.

Cuts

Any cat can get a cut, but it is especially likely with outdoor cats owing to fights, contact with machinery, and other hazards. First, stop any bleeding that may be present. This is usually quite easy to do with a pressure bandage, which is simply a pad of gauze held firmly against the cut with your hand. Severe bleeding can usually be controlled in the same way, while another person drives you and the injured cat to your veterinarian.

If the cut is a fairly fresh one—not more than a day old—it may heal best if stitched up. Many cuts, however, should not be stitched. This includes cuts that are dirty, cuts over joints, and cuts over heavily muscled areas. Often a bad-looking cut will heal completely in a month without suturing and without a scar. When the same wound is sewn up, healing takes about six weeks and often leaves a scar. One reason for this is that the outside of the cut heals before the inside, sealing in a few bacteria. These bacteria soon multiply and cause an infection. A very mild infection can retard healing for a lengthy period. So let your veterinarian make the decision about whether to suture that cut or not. Many veterinarians are nearly ordered to sew up a cut that they know will heal much better left open. People think the wound looks better if it is not gaping open, I guess. What they don't realize is that it would be around a shorter period of time if left open and treated with an antibiotic.

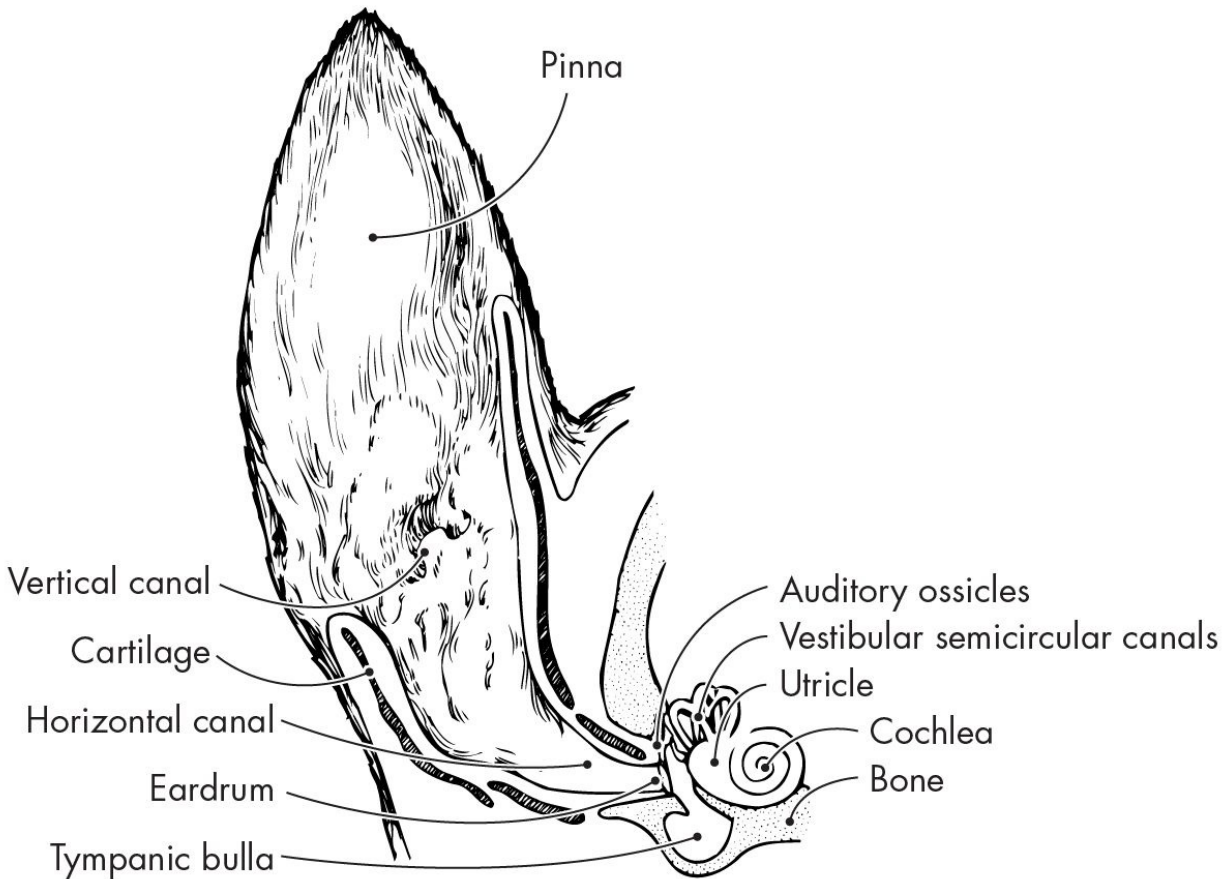
Bandages are another no-no in most cases. A bandage *can* be useful to hold a

pressure pack in place or to protect the animal from self-mutilation. Generally, though, a bandage retards healing by not allowing air to circulate freely around the wound.

Diarrhea

Diarrhea can be a disease or a *symptom* of a disease. Like humans, cats occasionally are bothered by diarrhea. Pampered house cats are most susceptible to this problem. Improper feeding is the most common cause of simple diarrhea. Feeding such things as greasy, spiced foods or excess milk or cream to a cat is just asking for gastric upsets.

When a cat is bothered by diarrhea, take away all of its food, but be sure to leave out a full water dish. Insufficient water intake along with diarrhea can cause dehydration. Give 1 teaspoonful of kaolin-pectin every 3 hours. If this does not stop the diarrhea in 12 hours or if the cat is running a fever or acting listless, call your veterinarian. Diarrhea can be a symptom of many illnesses, some serious. Feline distemper and severe parasitic infestations, for instance, often have an accompanying diarrhea.



The ear. When cleaning the ear, use only a cotton ball and your finger. Inserting a narrow object could easily injure the delicate structures inside the ear.

Ear Infection

Ear infections in cats are often brought about by ear mites. They burrow into the ear and provide places for bacteria and fungi to enter. This is often enhanced by the dirty, waxy accumulation present in the ear.

When the cat shakes its head, scratches at its ears with its hind feet, and yowls, it pays to check the ears. Sometimes there is just an accumulation of wax and dirt in the ear canal. This is easy to remedy by swabbing the ear, first with a ball of cotton moistened with alcohol or peroxide and then with a dry ball. Go down into the ear as far as you can reach with your finger. If you aren't too forceful and don't use anything smaller than a cotton ball and your finger, there's little chance you'll do any damage to the ear while cleaning it. This cleaning

may need to be repeated daily for two or three days.

If just cleaning the ear does not bring relief promptly, take your cat to the veterinarian for closer examination and proper diagnosis. Treating the ear for ear mites will not help a cat with a fungal or bacterial infection, which can be correctly diagnosed only by a veterinarian. Left untreated or treated improperly, ear infections can result in deafness or central nervous system conditions.

Feline Calicivirus

Feline calicivirus, often just called calicivirus, is a quite contagious respiratory disease of cats, especially kittens and debilitated cats. Its first symptoms include fever, nasal discharge, discharge from the eyes, ulcers of the tongue or nose, and lack of appetite. While the kitten or cat may be sick for quite a while, only the very young or severely stressed cat is apt to die. However, secondary bacterial infections are common, which may cause pneumonia or other serious problems.

Call your veterinarian as soon as you suspect your cat may have this disease. Because this problem is caused by a virus, treatment consists of controlling the symptoms, giving antibiotics to treat or prevent secondary infections, and very good nursing.

Prevention is always the best “treatment.” Have your cat vaccinated regularly, and reduce stress by providing good ventilation and keeping it away from dusty environments. Also, keep your cat away from sick cats that may be roaming around.

Feline Distemper

Feline distemper, or panleukopenia, is a very contagious viral disease. It may strike very suddenly, killing cats, especially kittens, before you notice any sign of illness. When symptoms are present, they include a fever, lack of appetite, diarrhea, vomiting, weakness, depression, and dehydration. This disease is often confused with poisoning as it strikes so quickly and has such a high mortality rate, sometimes wiping out an entire farm’s unvaccinated cat population in less than a week.

Treatment consists of administering antibiotics to prevent secondary infection, treating the diarrhea, giving intravenous electrolytes to treat and prevent further dehydration, and general good nursing. If the cat is running a subnormal temperature, placing it in a box on a heating pad often helps maintain a normal

temperature. Without the extra heat, the cat will usually die.

When the cat survives panleukopenia, gradually introduce it to mild foods in small amounts until it is eating normally. Isolate it from unvaccinated cats for two months because it may remain a carrier for that period. While treatment can be discouraging and sometimes expensive, vaccination is inexpensive and quite reliable.

Feline Infectious Peritonitis

Feline infectious peritonitis is a contagious feline disease caused by a virus, often resulting in death, even with treatment. Symptoms may be vague and can include a fever, diarrhea, respiratory distress, and sometimes conjunctivitis. Weakness and death usually follow.

Fortunately, this disease is not common; but with new vaccines available, your veterinarian may suggest vaccinating your cat, especially in areas where feline infectious peritonitis has been known to occur. If you suspect that your cat has this problem, contact your veterinarian right away.

Feline Leukemia

Feline leukemia is a serious viral disease, usually spread by contact with infected cats. The virus does not survive outside of the cat. Some cats may develop resistance to the virus after exposure, but some remain carriers, showing no sign of having the disease. Symptoms may include a lack of activity, thinness, anemia, and weakness. There may be tumors of many of the body organs, including the liver, lungs, and kidneys.

If you suspect feline leukemia, your veterinarian can test your cat to find out. There is no successful treatment, but knowing for sure and taking your veterinarian's advice may help protect any other cats you have at home.

It is safest to have your cat vaccinated against this killer. The cost is low, and it is worth the money.

Fishhooks in the Mouth

Cats, being naturally curious and also hunters, often get tangled up in fishing tackle. It's very easy for a sharp, barbed hook to get imbedded in the lips, but it is very hard to remove, particularly from the mouth of a frightened cat.

If only one hook is into the skin, it is sometimes possible to wrap the cat in a sturdy towel and clip the barbed point off with a pair of side cutters. An alternative is to squeeze the barb flat with pliers and then gently but firmly pull the hook.

If the cat is too frantic or there is more than one hook in the skin, it is best to take the cat to a veterinarian for the hook removal. He can give the cat an injection of quick-acting anesthetic that will immobilize it temporarily while the hooks are taken out. Often he will also give an injection of antibiotics and tetanus antitoxin to prevent later problems.

Fleas

Fleas are the main external parasites that bother cats. Lice are uncommon under most circumstances, and ticks are rarely a problem. (For more information on these latter two parasites, see “Lice” on page 323 and “Ticks” on page 329.)

Fleas on cats are quite easy to control if you find them before they infest your cat in large numbers. Dusting or spraying with a rotenone compound is safe and quite effective. Keep in mind that fleas do not stay on the cat—they can be outside in the sand, in your rugs, or in the cat’s bedding—so be sure to treat any suspect places at the same time you treat the cat. Remember that cats are extremely sensitive to many toxic insecticides; so read all labels very well, or you may poison your cat!

There are very effective topical products for cats, such as Frontline Plus and Bio Spot Spot On, which are applied to the skin of a cat, drawing a line from the shoulders to the tail. This drug is absorbed into the body and by biting the cat, the flea will be killed. It is also effective on any eggs. Be sure NOT to use a product labeled for use in dogs as cats are highly sensitive to many drugs and it could kill them or make them exhibit severe allergic reactions.

Also available are herbal flea collars, which repel fleas rather than killing them. There are seldom any adverse reactions, and the cat remains flea-free in most cases.

Fly Larvae

A cuterebra is a fly larva that is sometimes seen in cats, especially kittens. There is often only a tiny hole, in the neck area or under the chin. There may be a slight wetness in the area. The larva is large, sometimes nearly an inch long.

On close examination you can see the larva in a lump under the skin or moving around in the opening in the skin.

Never squeeze the larva as this can cause anaphylactic shock and kill the cat. Your veterinarian can carefully enlarge the opening slightly and remove the larva slowly with forceps. He will then flush out the wound and treat it with an antibiotic ointment or powder. Once the larva is removed, there are seldom any complications.

Hair Balls

Hair balls are a common problem in cats, especially longhaired cats. As the cat cleans itself, it often swallows loose hairs. In the stomach these hairs can mass together and form a wad, which may be as large as 6 to 8 inches long × 2 to 3 inches in diameter. The cat may vomit these wads or show gastric distress from their presence. When passed further down the digestive tract, they can cause constipation or even complete intestinal blockage in some instances.

Commercial hair ball preparations with a palatable base are useful in helping your cat expel hair balls before they become a serious problem. Mineral oil, milk of magnesia, and castor oil may also bring relief, but they are hard to administer. Giving an oily canned food, such as tuna or salmon, at least twice a week helps some cats.

The best remedy for hair balls is prevention. Daily brushing, which removes a good portion of shed hairs as they occur, will do a lot in relieving hair ball problems in your cat.

Hypocalcemia

Hypocalcemia occurs in mother cats following the birth of kittens, usually when they are from one to three weeks of age. It is most common where there is a very large litter of healthy, hungry kittens and the mother is a good milk producer. The queen needs to produce immense quantities of milk to supply these kittens, and she draws the calcium from her blood. As her blood calcium level falls, she becomes uncoordinated, stiff, shaky, and nervous. She may have seizures or go into a coma.

Fortunately, hypocalcemia isn't common in cats. But when this problem does occur, your cat must get veterinary treatment soon after symptoms appear, or she will die. The veterinarian will give an intravenous injection of calcium to replace

the depleted calcium in her blood. Recovery is dramatic and fast. The kittens should be weaned or bottle-raised as allowing them to continue nursing will usually cause the same condition to recur, and the next time it may kill her.

Mange

Mange is not often seen in cats, but it does occur. Mange is caused by several different types of mites that burrow under the skin, but the results are about the same. The first signs of mange are usually scabby spots, often around the face, where the hair suddenly has fallen off. There may be a foul discharge, intense itching, thickened skin, large crusty areas, and large areas of hair loss.

If you suspect your cat has mange, have your veterinarian do an examination. Several other problems can resemble mange, and you'll need a correct diagnosis to choose the right treatment. Also, the earlier you discover and treat the condition, the quicker and more effective the treatment will be.

Clipping the hair short over the entire affected area will help improve the effectiveness of the treatment. Since the mites burrow under the skin, they are difficult to treat with common insecticides, such as those used for fleas. And cats are especially sensitive to toxic compounds, so it is sometimes difficult to treat them effectively. Your veterinarian may use a shampoo containing lime-sulfur or rotenone as well as lotions or ointments containing these ingredients. *Do not use a mange remedy formulated for dogs, because cats are very susceptible to many chemicals.* Many veterinarians have had good luck using Ivermectin or Revolution in treating cats for mange. However, neither product is labeled for use in cats. But under close veterinary supervision, they can sometimes work wonders when other treatments fail. Also, take great care to prevent the cat from licking off the mange remedy; use an Elizabethan collar. (See "Restraint" on page 337 and the illustration on page 339 for instructions on making and using a collar.) Often, repeated treatments are needed to completely rid the cat of mange mites.

Mastitis

Mastitis is an inflammation of one or more of the queen's mammary glands. It is often caused by a bacterial infection that invades the gland due to stress. Some common types of stress include injuries (such as bruises, falls, being stepped on, cat bites, or getting banged in a door), copious milk supply not taken by kittens, and wounds to the teats.

When you first notice the problem, the cat may act sick or in pain, and she may have a fever. The affected gland is often swollen and hard, and there is usually heat in the area of infection. The kittens may also be affected by abnormal milk, acting listless or mewling. Remove them to a warm nest right away, and keep them off the mother until she is better. (You can bottle-feed them, if necessary. For more information on this technique, see “Orphan Kittens” on page 342.)

Consult your veterinarian as soon as you suspect mastitis. Treatment often consists of giving injections of a broad-spectrum antibiotic, applying hot packs on the affected area, massaging the area, and gently squeezing as much milk as possible from the affected glands several times during the day.

Phenol Poisoning

Cats are highly susceptible to phenol and phenol compounds as well as to iodine. Many household cleaners contain phenol, and many cats are poisoned unintentionally when drinking out of a toilet bowl containing Lysol or another cleaner. Never use a phenol shampoo, such as those often used on dogs, on a cat. If you mop your floor with a cleaner that contains phenol, rinse it well and dry it before allowing the cat into the room. Licking wet paws containing a heavy phenol concentration can cause serious, and sometimes irreversible, gastric upsets.

A cat that has ingested large amounts of phenol will often sustain extensive damage before you discover the illness. Successful treatment is very difficult in these cases. When smaller amounts are ingested, inducing vomiting, providing intravenous fluids and preparations to soothe the stomach, and good aftercare will often save the cat. If you suspect phenol poisoning, take your cat to the veterinarian at once. A few hours may make the difference between successful treatment and disappointment.

Pneumonitis

Pneumonitis is caused by a highly contagious virus. It is spread by direct contact between infected cats and unvaccinated healthy cats. The first signs you'll notice are usually sneezing and coughing. There is often a discharge from the eyes and nose. The cat generally runs a fever. Kittens often have a weight loss as they lose their appetite. They are generally more severely affected by the disease than are older cats.

It is important to remember that there are other feline diseases that have similar symptoms, so be sure to contact your veterinarian at once for a correct diagnosis. Treatment includes intravenous electrolytes, a broad-spectrum antibiotic to prevent secondary infections, such as pneumonia, and good general nursing. Treated fairly early in the course of the disease, most cats recover with little trouble, although they may retain the infection and become sick again later in life if stressed.

Vaccination is very effective and inexpensive protection for your cats. (Your veterinarian may use the term *chlamydia*, which is often a causative agent.)

Rabies

Rabies is not as common in cats as it is in bats, skunks, raccoons, foxes, and dogs, perhaps owing to the cautious nature of cats. But, unfortunately, it can and does occur. It is always fatal once signs of the disease appear. And, of course, it can be spread to humans with fatal results unless treatment is given immediately, before symptoms occur.

Rabies is spread most often by a bite contaminated by saliva, which carries the virus. Exposure is possible from other routes, but it is uncommon. Cats frequently get rabies by catching or being bitten by rabid bats.

Many rabid animals develop a complete personality change. A quiet, loving cat will become shy or vicious. A nasty cat will suddenly become loving and want to be cuddled. There is often a change in the voice due to paralysis of the throat. The rabid cat will often either quit eating or develop a depraved appetite (eating dirt, stick, rocks, or other abnormal items). There may be stiffness in the hind limbs.

There are two types of rabies: furious rabies and dumb rabies. With furious rabies fewer people are actually endangered because they naturally avoid an aggressive, biting, scratching cat. But a cat that has dumb rabies just sits around with its lower jaw hanging down, usually with saliva running out of its mouth. Owners often think that the cat just has a bone stuck in its mouth, and they become infected when they get scratched on the teeth and get the disease-spreading saliva rubbed into the abrasion.

If you see a cat with these symptoms, treat it with caution. Isolate it in an escape-proof carrier, and take it to your veterinarian at once. (Don't, however, think a female in heat has rabies because of her "change in voice" and strange behavior!) For more information on handling rabies, see "Rabies" on page 327.

There is no treatment for rabies in animals, but there is a very effective vaccine available to protect both your cat and your family. (Remember that often a cat is the link between possibly rabid wildlife and your family.) Be sure to keep your cat's vaccination current.

Rhinotracheitis

Rhinotracheitis resembles pneumonitis as the cat has mattered eyes and a copious discharge from both nose and eyes. It coughs, sneezes, and runs a fever. Like pneumonitis, rhinotracheitis is caused by a very contagious virus, frequently complicated by an additional secondary bacterial infection. Except in kittens, the disease is serious but seldom fatal.

Consult your veterinarian right away if you suspect your cat has this problem. Treatment consists of treating the symptoms, giving antibiotics to treat or prevent secondary infections, fluids, and good nursing.

As with most viral infections, vaccination is much more satisfactory than trying to treat the disease. Good combination vaccinations are now available for the major respiratory infections of cats, so it's easy to prevent problems.

Ringworm

Ringworm is caused by a fungus, often *Microsporum canis*. It is not a worm, nor is it caused by a worm. It is a circular lesion, often appearing quite suddenly. The skin is commonly rough, with the hair gone from the spot. If left untreated, the lesion gets larger and larger, usually retaining the circular pattern.

Some forms of ringworm are contagious to humans. All can be passed back and forth among animals of a like species. So if you suspect ringworm, isolate the affected cat immediately, and then check all the other cats or kittens right away. Take any affected cat to your veterinarian for examination. It is a good idea to wear gloves when handling a cat with ringworm.

Treatments vary from soaking the cat in a medicated shampoo to treating just the spots with a low-toxicity fungicide. (Iodine preparations are often used in treating ringworm in cattle and dogs, but most cats are highly sensitive to iodine, and they have *severe* trouble if you use it on them.) If you treat the spots at home, be sure to use rubber gloves and apply the medication from the outside of the spot to the inside of the spot; working outward may spread the fungus.

If you own a number of cats and want to prevent this problem, there is a

vaccination available for ringworm in cats. Ask your veterinarian for his opinion on its use in your situation.

Urinary Calculi

Urinary calculi trouble male cats more often than females due to the size of the passageway through which the urine exits the body. There is little agreement on whether or not castration affects the incidence of the problem. Some studies have been done, and they reveal no significant difference in the size of the urethra between neutered and unneutered males; but many veterinarians do believe there is a definite connection between castration of young males and urinary calculi. For this reason, it's generally recommended that you wait until the male is seven to nine months old to castrate it.

Feeding a diet of only dry food will often contribute to problems with urinary calculi. When fed a moist diet, such as canned food or meat, the cat gets additional moisture through the food. When eating dry food, the cat often does not drink enough additional water.

A change in the pH of the bladder, whether due to an infection or other causes, can also contribute to urinary calculi. As the pH in the bladder becomes more alkaline, salts, crystals, and calcium precipitate out, forming the calculi.

Diets high in ash and magnesium have been shown to cause a high incidence of urinary calculi—perhaps the greatest cause of calculi problems in cats.

Many times the cat has passed several smaller calculi (sometimes called “sand”) before one lodges and blocks the urethra. Often the first symptoms you'll notice are straining at the litter tray with scanty results, usually with increased frequency. Dullness and loss of appetite soon follow. You can feel the distended bladder as a hard, roundish bulge in the abdomen. (Do not squeeze the bladder, because it may rupture, usually causing the death of the cat.) Take the cat to the veterinarian immediately for treatment. As the problem progresses, the cat becomes toxic from absorbing the waste material. When it becomes very ill from uremia, treatment is very difficult, and the cat often dies.

Treatment in severe cases consists of emptying the bladder, by use of a catheter (which passes through the urethra) or a direct tap (a needle inserted through the abdomen into the bladder), and flushing the urethra to dislodge the plug. Once the cat is out of danger and urinating on its own, you must place it on a strict diet that is low in magnesium and ash. Many appropriate foods are available commercially; your veterinarian can recommend one to you. Or you

may make up your own, following his prescribed diet of homemade ingredients. Remember, though, that once a cat has had urinary calculi, placing it on regular commercial food again will usually result in recurring attacks of the problem.

Warfarin Poisoning

Warfarin is the principal ingredient of several popular rodent poisons. Even though it is supposedly safe for children and pets, cats can be poisoned by it. This most often happens when the cat eats the rodent bait directly or eats a large rodent with a stomachful of fresh bait. Once the bait has passed through the digestive tract and kills the rodent, it usually will not harm a cat, even if the cat eats the rodent.

Warfarin and other rodenticides contain a drug, dicoumarin, which causes hemorrhage throughout the body. Affected cats seem weak, anemic, and depressed. They do not appear to be in pain or severely distressed.

If you suspect that your cat may have had access to a rodent poison containing dicoumarin, take the cat immediately to your veterinarian. In most cases injections of vitamin K and confinement will aid in helping the blood begin to clot. In severe cases blood transfusions and fluids are also given.

Never leave rodent poison where it is at all possible for pets or children to reach it. Place it in a sturdy box, accessible only to rodents, and set it far away from pet feeding places or children's play areas.

Worms and Internal Parasites

Although cats are not severely bothered by worms as often as dogs are, there are several types that do infest them on occasion.

Hookworm

Hookworm is common in cats. These small worms are not seen in the stool, nor are they normally visible with the naked eye. Symptoms include anemia, bloody stools, diarrhea, and unthriftiness. Young kittens are the hardest hit and often die suddenly.

Roundworm

Roundworm is the most common cat parasite. This worm is visible to the

naked eye and looks like a bean sprout. It is most often seen after being vomited up, often along with a hair ball. A severely infested cat will have a potbelly but remain thin despite a good diet. It will also have a poor hair coat and act listless.

Tapeworm

Tapeworm, which quite often infests cats, is carried into the cat via the flea. While cleaning itself, the cat can swallow a flea containing tapeworm larvae. Soon the cat is infested with an adult tapeworm, which can be up to 2 feet long. Cats can also get tapeworm through eating infested rabbits or mice or other rodents. Tapeworm segments are eliminated with stool and often cling to the rectal area. When just passed, they look like inchworms, only white; when dry, they look like small grains of rice. A cat with tapeworm may just appear a bit unthrifty or have digestive or nervous disorders. Some cats do not show signs of trouble but still are infested with tapeworms.

Other Parasites

There are many other worms and intestinal parasites that can and do infest cats. A few of them are threadworms, lung flukes, liver flukes, *Giardia*, coccidia, lungworms, heartworms, and *Babesia felis*. Many come from eating infested rodents. Others are picked up when the cat licks the eggs off its feet while washing. A single indoor cat is much less likely to pick up worms than is a cat from a cattery, an outdoor cat, or a cat that goes out where stray cats and other animals roam.

Treating Parasites

When you suspect your cat has intestinal parasites, be sure to take a fecal sample to your veterinarian for examination. Each worm or group of worms has one or two medications that work best on it, and some worms require rather complicated treatments. Using a “shotgun” wormer from the drugstore will do your cat no good and may even be dangerous, should the cat be severely debilitated or sick from a problem other than worms.

Every cat should have a regular fecal examination for worms and other intestinal parasites at least once a year. This examination is simple, quick, and inexpensive, and it’s worth every penny. With many parasites you will not see visible damage until there’s not much you can do to save the animal. Lungworms, for instance, cause scarring and infections in the respiratory tract,

while many protozoa (one-celled organisms) cause ulceration in the intestinal tract, which can become infected. Many worms cause digestive upsets, sometimes severe enough to kill the cat, as they burrow into the mucous lining of the stomach and intestines. Hookworm can cause anemia, which can bring about sudden death. And large migrating worms or dead adult worms can cause plugs in blood vessels or heart attacks. The point is that the few minutes it takes to collect a small fecal sample and take it to your veterinarian could easily save your cat's health, if not its life.